

Malware Analysis

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 VirusTotal



Agenda

- Introduction about malware and its types
- Malware analysis process
- Project goal and requirements
- Steps for projects:
 - Anti-malware scanning and find the signature for sample malware using virus total
 - Finding the string values of sample malware using PEViewer tool
 - Malware is packed or unpacked with packed malware form information using PEiD
 - Finding malware implementation and date using PE-Explorer
 - Finding disassembler result using PE-Explorer
 - List of Import functions using Dependency scanner option of PE-Explorer
- Conclusion
- Ref.
- Thank You



Warning



- The information provided in this presentation is only intended for educational purpose only.
- Any collected data during testing must be handled with strict confidentiality and used only for the purposes outlined in this project.
- Unauthorized scanning or exploitation of system is illegal and unethical



Introduction about Malware and its types

The objective of this project is to perform malware analysis on sample malware files using tools like virus total, PEViewer Tools, Pe-Explorer, PEID understanding and defending against these diverse threats.

Malware (short for malicious software) is any program or code designed to harm, exploit, or disrupt systems, networks, or devices. Cybercriminals use malware to steal sensitive data, gain unauthorized access, sabotage systems, or monitor user activity. Malware is a significant threat in the cybersecurity domain, requiring robust defense and analysis techniques to mitigate its impact.

Cont. Introduction about Malware and its types

Types of Malware

Viruses: Self-replicating programs that infect files and spread when executed

Worms: Standalone programs that spread across networks without user intervention.

Trojan Horses: Malicious software disguised as legitimate applications to trick users into installation.

Ransomware: Encrypts data and demands payment to restore access

Spyware: Secretly monitors user activity and collects sensitive information

Keyloggers: Record keystrokes to steal credentials and sensitive data.

Malware Analysis process

Anti-malware Scanning:

- Perform an initial scan of the sample malware using **Virus Total** to identify its signature and check for known threats across multiple anti-virus engines

String Analysis:

- Use the **PEViewer** tool to extract **string values** from the malware sample. This can reveal hardcoded file paths, IP addresses, or malicious commands embedded in the executable.

Packed or Unpacked Analysis:

- Check if the malware is packed using **PEiD**. If packed, note the packer type and consider unpacking the malware for deeper analysis.

Cont. Malware Analysis process

Metadata Inspection:

- Analyze the **implementation date** and other metadata using **PE-Explorer** to determine when and how the malware was created.

Disassembly:

- Use the **Disassembler** feature in **PE-Explorer** to reverse-engineer the malware and inspect its low-level code. This helps in understanding its behavior and potential exploits.

Import Functions Analysis:

- List all **imported functions** using the **Dependency Scanner** in **PE-Explorer**. This step identifies external libraries or APIs that the malware relies on for its malicious activities.

Project goals and requirements

This project will be done in a Virtual environment network and machine

The primary system is Windows server 2022.

Project Goals:

- Understand Malware Behavior
- Identify Malware Signatures:
- Analyze Malware Structure
- Extract Import Functions
- Evaluate Disassembly Results
- Enhance Threat Response

Tools used:

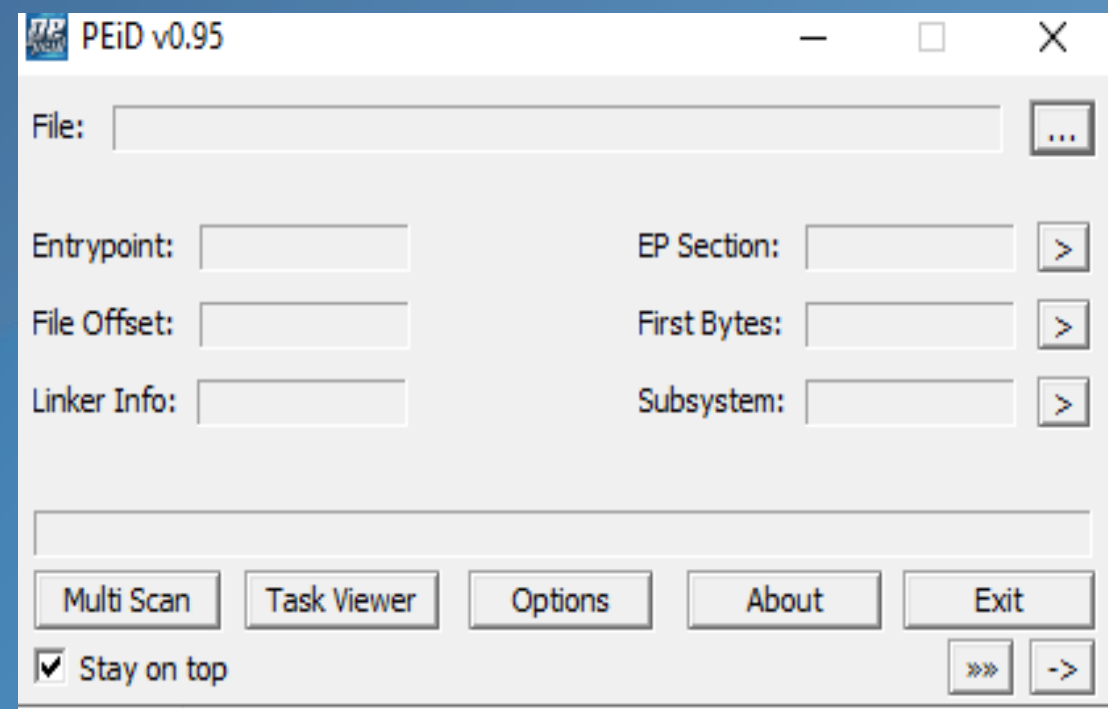
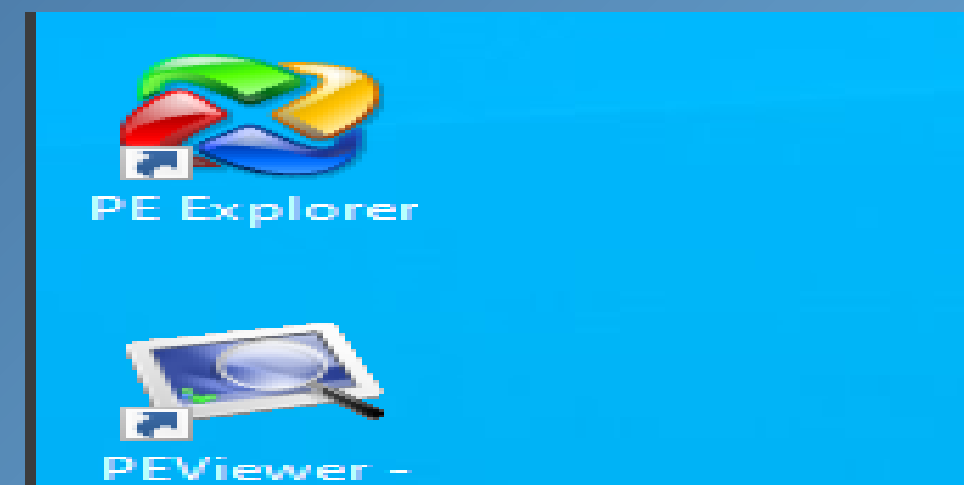
- Virus Total
- PEViewer
- PEiD
- PE-Explorer

Sample Malware:

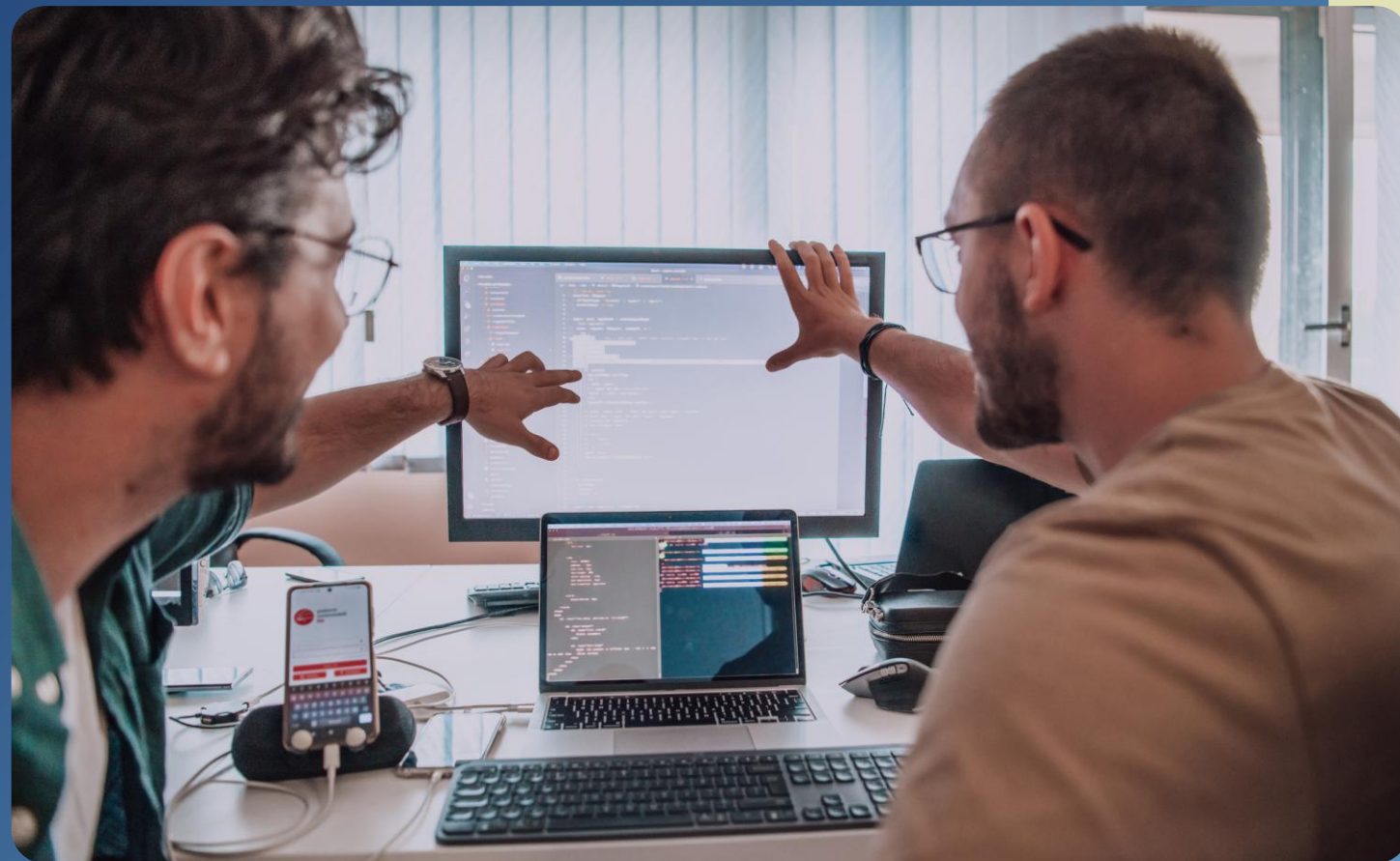
Lab01:

Lab02-04

Cont. Project goals and requirements



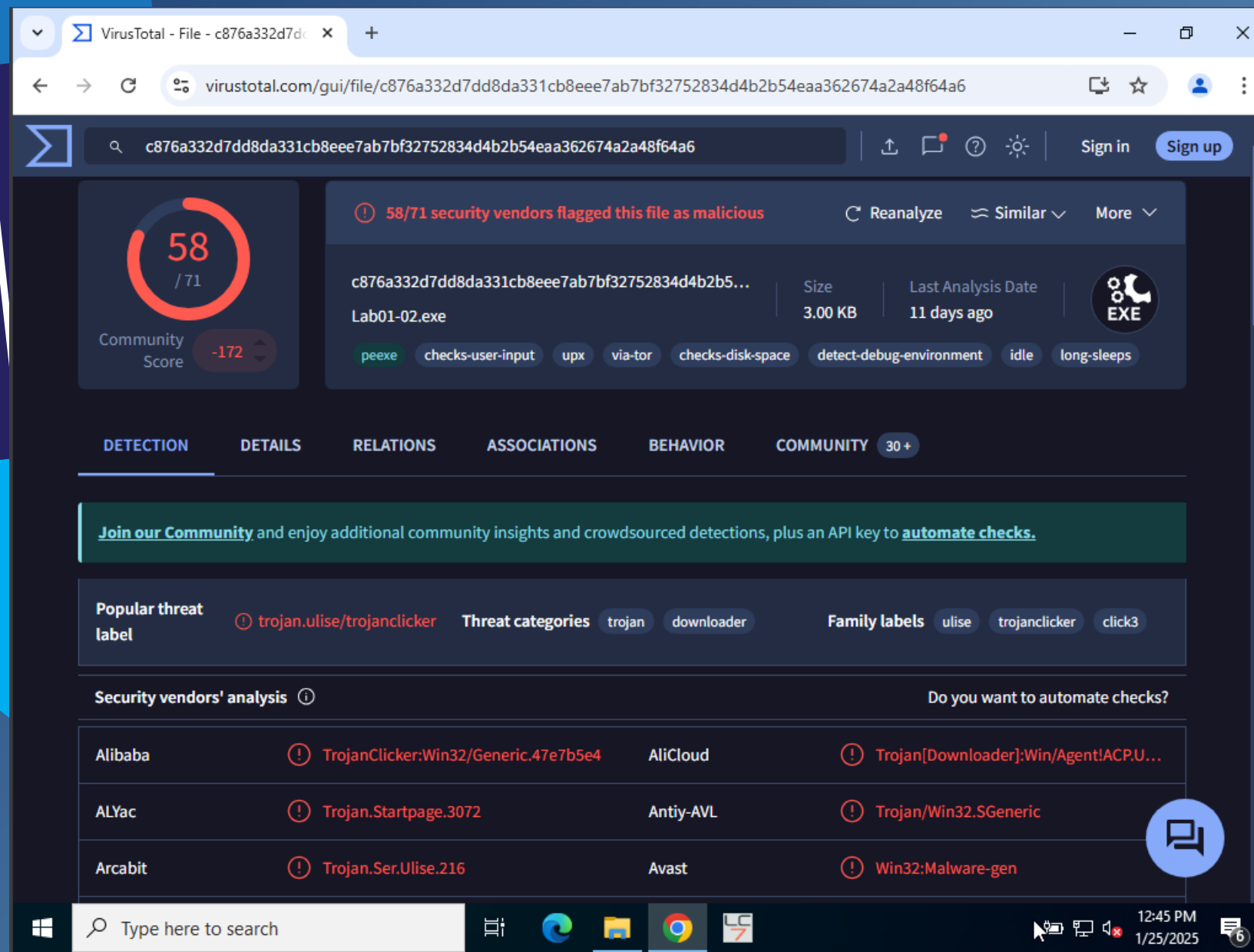
Steps for Project



1. Anti-malware scanning and find the signature for sample malware using virus total
2. Finding the string values of sample malware using PEViewer tool
3. Malware is packed or unpacked with packed malware form information using PEiD
4. Finding malware implementation and date using PE-Explorer
5. Finding disassembler result using PE-Explorer
6. List of Import functions using Dependency scanner option of PE-Explorer

Step 1: Lab01-02

All the screenshots in Step 1 for the sample malware Lab02-04 is the result of the anti-malware scan and the signature associated with the virus using virus total.



The screenshot shows the VirusTotal interface for a file named 'Lab01-02.exe'. The file has been analyzed by 58 out of 71 security vendors, with 172 community flags. The file is identified as a trojan (TrojanClicker) and is associated with the threat 'TrojanClicker.Win32/Generic.47e7b5e4'. The file size is 3.00 KB and it was last analyzed 11 days ago. The file type is EXE. The file is associated with the threat 'TrojanClicker.Win32/Generic.47e7b5e4'.

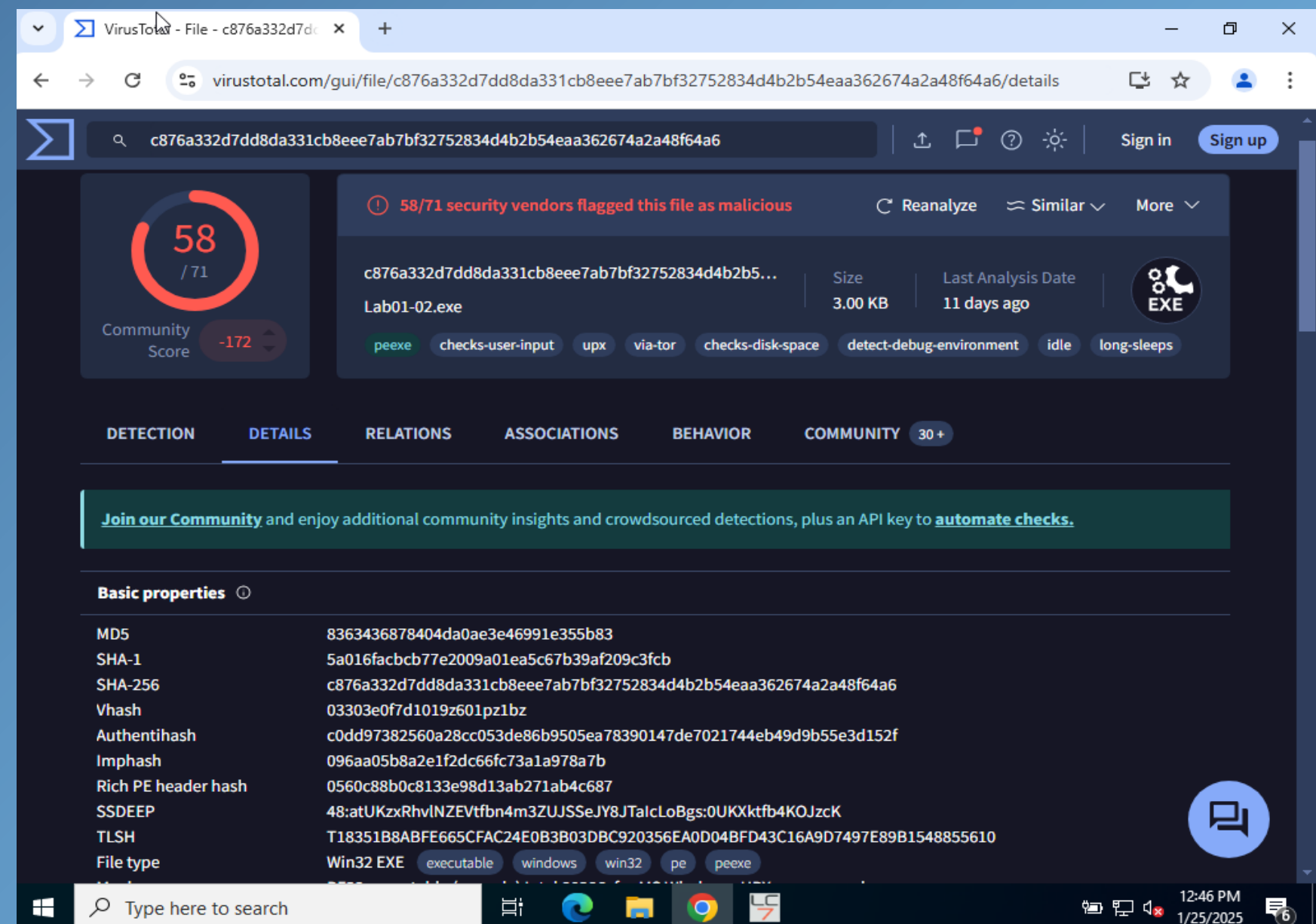
Community Score: 58 / 71

Threat categories: trojan, downloader

Family labels: ulise, trojanclicker, click3

Security vendors' analysis:

Vendor	Detection	Signature
Alibaba	TrojanClicker.Win32/Generic.47e7b5e4	AliCloud
ALYac	Trojan.Startpage.3072	Antiy-AVL
Arcabit	Trojan.Ser.Ulise.216	Avast



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ALYac	Trojan.Startpage.3072	Antiy-AVL
Arcabit	Trojan.Ser.Ulise.216	Avast

Basic properties:

Property	Value
MD5	8363436878404da0ae3e46991e355b83
SHA-1	5a016facbcb77e2009a01ea5c67b39af209c3fcb
SHA-256	c876a332d7dd8da331cb8eee7ab7bf32752834d4b2b54eaa362674a2a48f64a6
Vhash	03303e0f7d1019z601pz1bz
Authentihash	c0dd97382560a28cc053de86b9505ea78390147de7021744eb49d9b55e3d152f
Imphash	096aa05b8a2e1f2dc66fc73a1a978a7b
Rich PE header hash	0560c88b0c8133e98d13ab271ab4c687
SSDEEP	48:atUKzxRhvINZEVtfn4m3ZUJSSeJY8JTalcLoBgs:0UKXktfb4KOJzcK
TLSH	T18351B8ABFE665CFAC24E0B3B03DBC920356EA0D04BFD43C16A9D7497E89B1548855610
File type	Win32 EXE

Step 1: Lab01-03

VirusTotal - File - 7983a582939924c70e3da2da80fd3352ebc90de7b8c4c427d484ff4f050f0aec

virustotal.com/gui/file/7983a582939924c70e3da2da80fd3352ebc90de7b8c4c427d484ff4f050f0aec

Max size 650MB

7983a582939924c70e3da2da80fd3352ebc90de7b8c4c427d484ff4f050f0aec

66 / 72

Community Score -183

66/72 security vendors flagged this file as malicious

Reanalyze Similar More

7983a582939924c70e3da2da80fd3352ebc90de7b8c4c427d484ff4f050f0aec

Lab01-03.exe

Size 4.64 KB

Last Analysis Date 1 month ago

EXE

peexe fsg long-sleeps overlay detect-debug-environment direct-cpu-clock-access runtime-modules checks-user-input via-tor

DETECTION DETAILS RELATIONS ASSOCIATIONS BEHAVIOR COMMUNITY 30+

Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.

Popular threat label trojan.graftor/genome Threat categories trojan spyware Family labels graftor genome trojanclicker

Security vendors' analysis Do you want to automate checks?

AhnLab-V3	Trojan/Win.Generic.R427327	Alibaba	TrojanClicker.Win32/Tnega.79cba6fb
AliCloud	Trojan:Win/Graftor.GX22XJC	ALYac	Gen:Variant.Graftor.968808
Antiy-AVL	Trojan/Win32.SGeneric	Arcabit	Trojan.Graftor.DEC868
Avast	Win32:Evo-gen [Trj]	AVG	Win32:Evo-gen [Trj]

VirusTotal - File - 7983a582939924c70e3da2da80fd3352ebc90de7b8c4c427d484ff4f050f0aec

virustotal.com/gui/file/7983a582939924c70e3da2da80fd3352ebc90de7b8c4c427d484ff4f050f0aec/details

Max size 650MB

7983a582939924c70e3da2da80fd3352ebc90de7b8c4c427d484ff4f050f0aec

66 / 72

Community Score -183

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DETECTION DETAILS RELATIONS ASSOCIATIONS BEHAVIOR COMMUNITY 30+

Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.

Basic properties

MD5	9c5c27494c28ed0b14853b346b113145
SHA-1	290ab6f431f46547db2628c494ce615d6061ceb8
SHA-256	7983a582939924c70e3da2da80fd3352ebc90de7b8c4c427d484ff4f050f0aec
Vhash	04303d0d7d1bz2lz
Authentihash	b12ac29217ded1aacef00c41d707eec62d429519d67138ef0ded9834c2cd98b1
Imphash	87bed5a7cba00c7e1f4015f1bdae2183
SSDEEP	24:erDeoULXQeWKPUA4FOopcBl+PxYhIWlsp97IGg4QQL5ACqk22:GeoULAevPUA007vL8spDGnNLW7g
TLSH	T198A1A7262BF71DB0DC04577A90CC488AD1BD297596F3DE414D4D1567BDF86841C70E43
File type	Win32 EXE executable windows win32 pe peexe
Magic	PE32 executable (console) Intel 80386, for MS Windows
TrID	Win16 NE executable (generic) (28.5%) Win32 Executable (generic) (25.4%) Win16/32 Executable Delphi generic (11.7...)

6 new notifications

Step 1: Lab01-04

VirusTotal - File - 0fa1498340fca6c562cfa389ad3e93395f44c72fd128d7ba08579a69aaf3b126

virustotal.com/gui/file/0fa1498340fca6c562cfa389ad3e93395f44c72fd128d7ba08579a69aaf3b126

0fa1498340fca6c562cfa389ad3e93395f44c72fd128d7ba08579a69aaf3b126

63 / 72

Community Score -136

63/72 security vendors flagged this file as malicious

Reanalyze Similar More

0fa1498340fca6c562cfa389ad3e93395f44c72fd128d7ba08579a69aaf3b126

Lab01-04.exe

Size 36.00 KB

Last Analysis Date 1 day ago

EXE

peexe armadillo checks-user-input idle via-tor

DETECTION DETAILS RELATIONS ASSOCIATIONS BEHAVIOR COMMUNITY 30+

Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.

Popular threat label trojan.cerbu/gofot Threat categories trojan downloader droppe Family labels cerbu gofot dlder

Security vendors' analysis Do you want to automate checks?

AhnLab-V3	Trojan.Win.DownLoader.C5520690	Alibaba	TrojanDownloader:Win32/Gofot.7e5f679f
AliCloud	Trojan[downloader]:Win/Gofot.gyf	ALYac	Gen:Variant.Cerbu.64782
Antiy-AVL	Trojan[Downloader]/Win32.AGeneric	Arcabit	Trojan.Cerbu.DFD0E
Avast	Win32:DropperX-gen [Drp]	AVG	Win32:DropperX-gen [Drp]

VirusTotal - File - 0fa1498340fca6c562cfa389ad3e93395f44c72fd128d7ba08579a69aaf3b126

virustotal.com/gui/file/0fa1498340fca6c562cfa389ad3e93395f44c72fd128d7ba08579a69aaf3b126/details

0fa1498340fca6c562cfa389ad3e93395f44c72fd128d7ba08579a69aaf3b126

63 / 72

Community Score -136

63/72 security vendors flagged this file as malicious

Reanalyze Similar More

0fa1498340fca6c562cfa389ad3e93395f44c72fd128d7ba08579a69aaf3b126

Lab01-04.exe

Size 36.00 KB

Last Analysis Date 1 day ago

EXE

peexe armadillo checks-user-input idle via-tor

DETECTION DETAILS RELATIONS ASSOCIATIONS BEHAVIOR COMMUNITY 30+

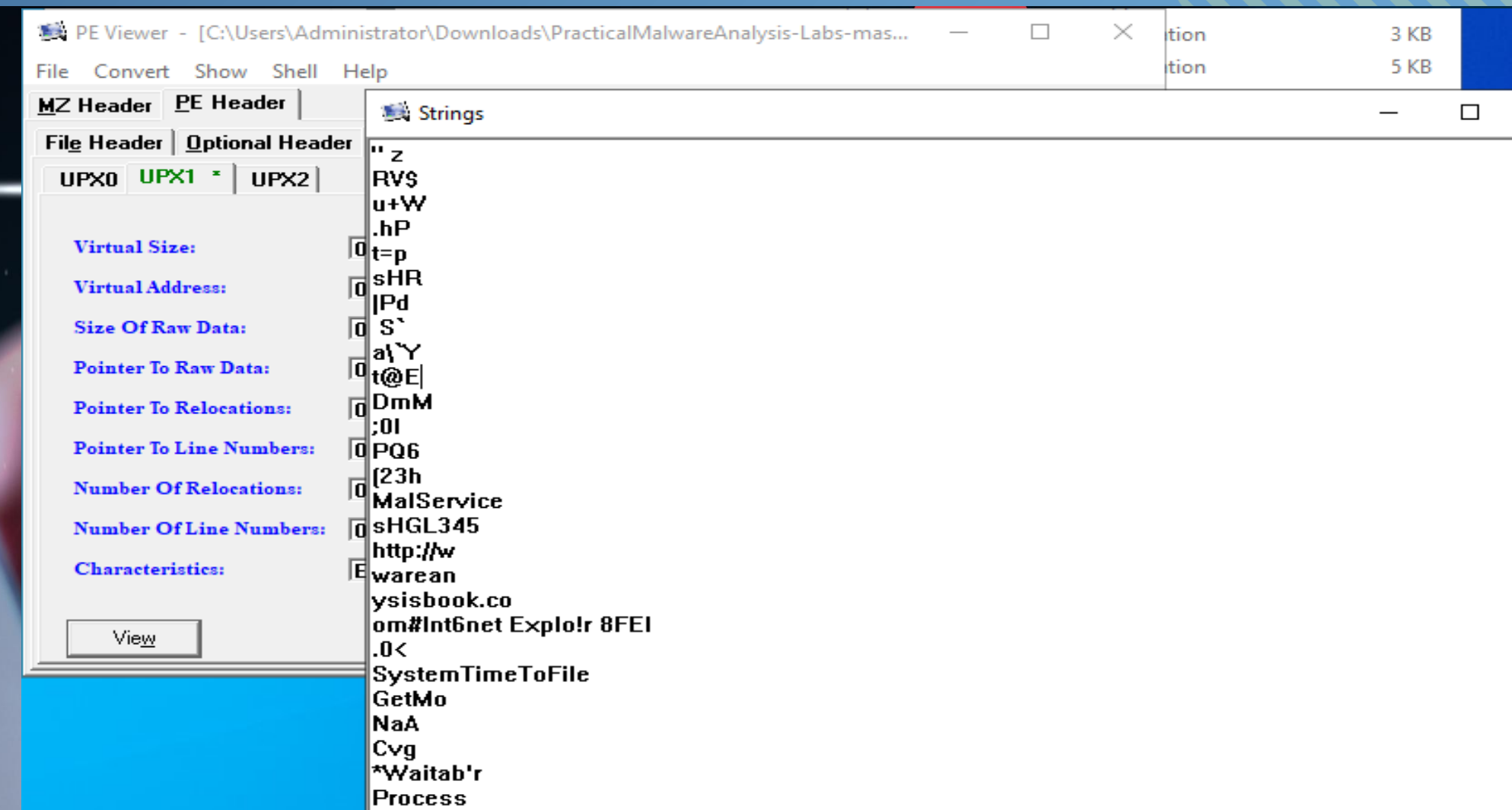
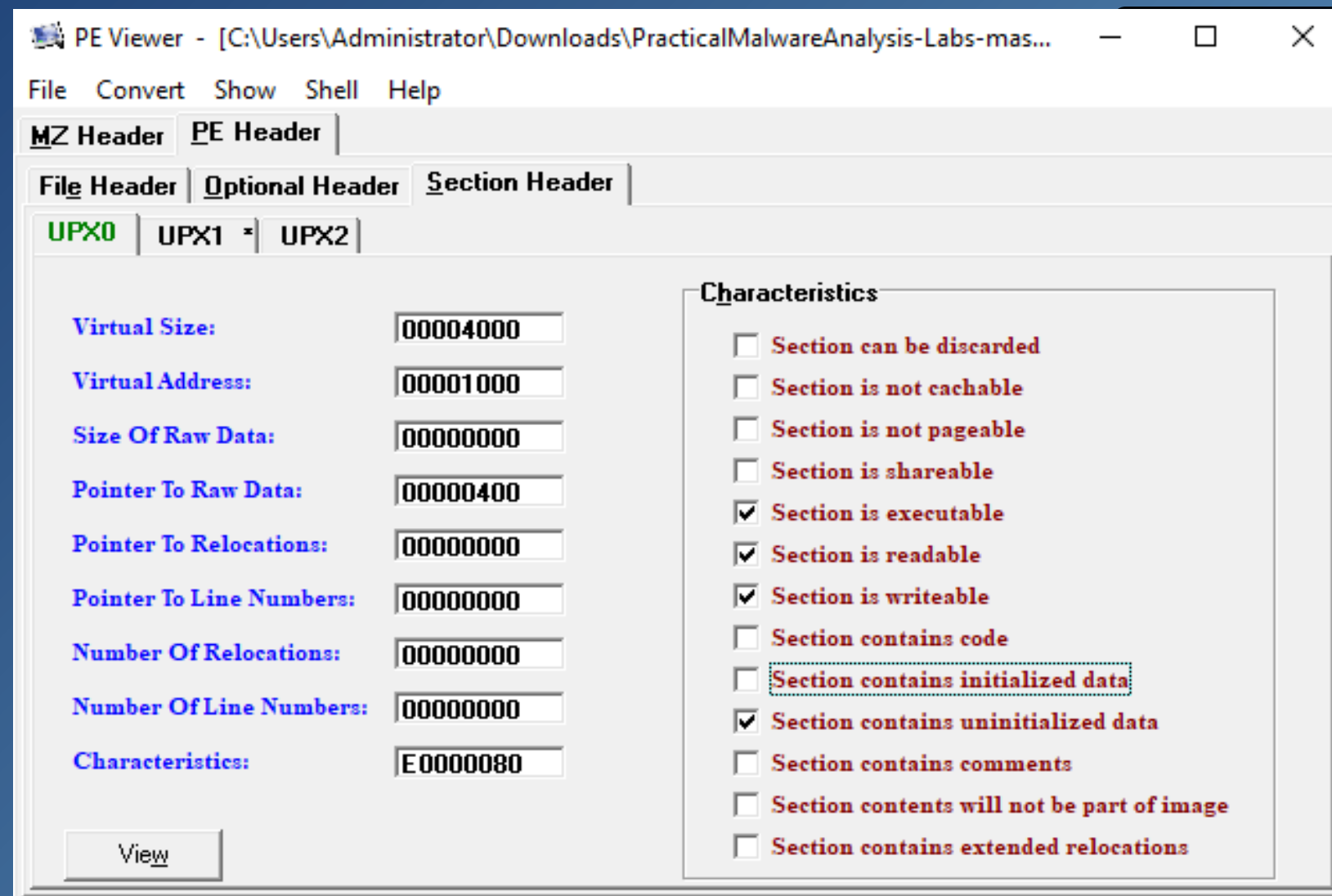
Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.

Basic properties

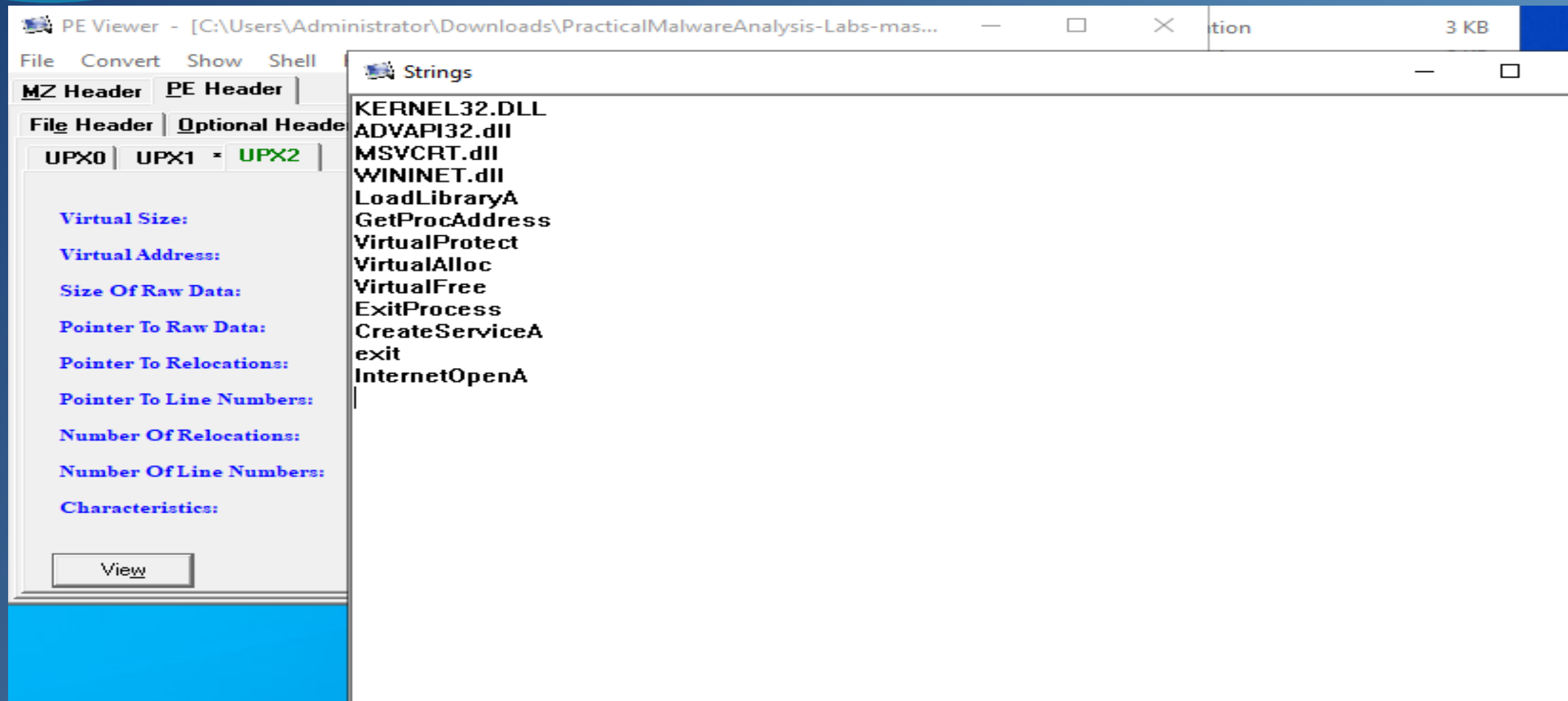
MD5	625ac05fd47adc3c63700c3b30de79ab
SHA-1	9369d80106dd245938996e245340a3c6f17587fe
SHA-256	0fa1498340fca6c562cfa389ad3e93395f44c72fd128d7ba08579a69aaf3b126
Vhash	034046151d151038z100f=z
Authentihash	e4d9d8ea008b5521c4b4273b8a276cf618db3f8af0bdd2f17d50f6c09e5bc150
Imphash	aade0ea6fbdcd9b8e96fe999cae6f603
Rich PE header hash	a9ce8adbba583f6837fc888ad6f8789e
SSDEEP	96:TF0MgAr71nxY9AAIvqZ2ZNNHsP4oyLnKcm5OzG38U6p2WL4P4oyn:JJaPLJC2ZNNHMP4oyLnKl38jp2VP4oyn
TLSH	T14EF2A7476B14D432D7884176262F82E68713697213B941CF9BF7568C85B6CE3923EF07
File type	Win32 EXE executable windows win32 pe peexe
Magic	PE32 executable (GUI) Intel 80386, for MS Windows
TrID	Microsoft Visual C++ compiled executable (generic) (38.4%) Win32 Dynamic Link Library (generic) (15.3%) Win16 NE e...

Step 2: Lab01-02

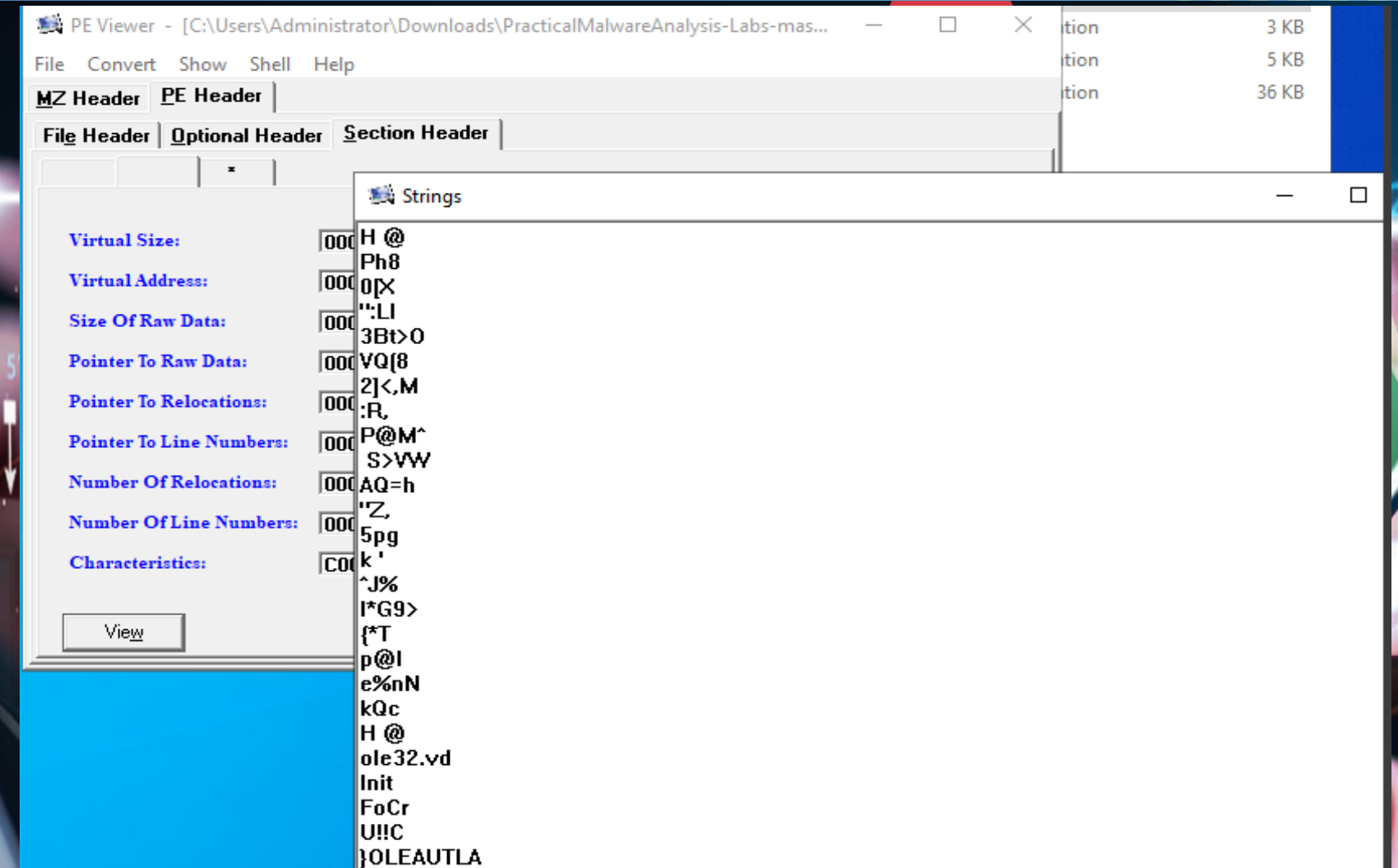
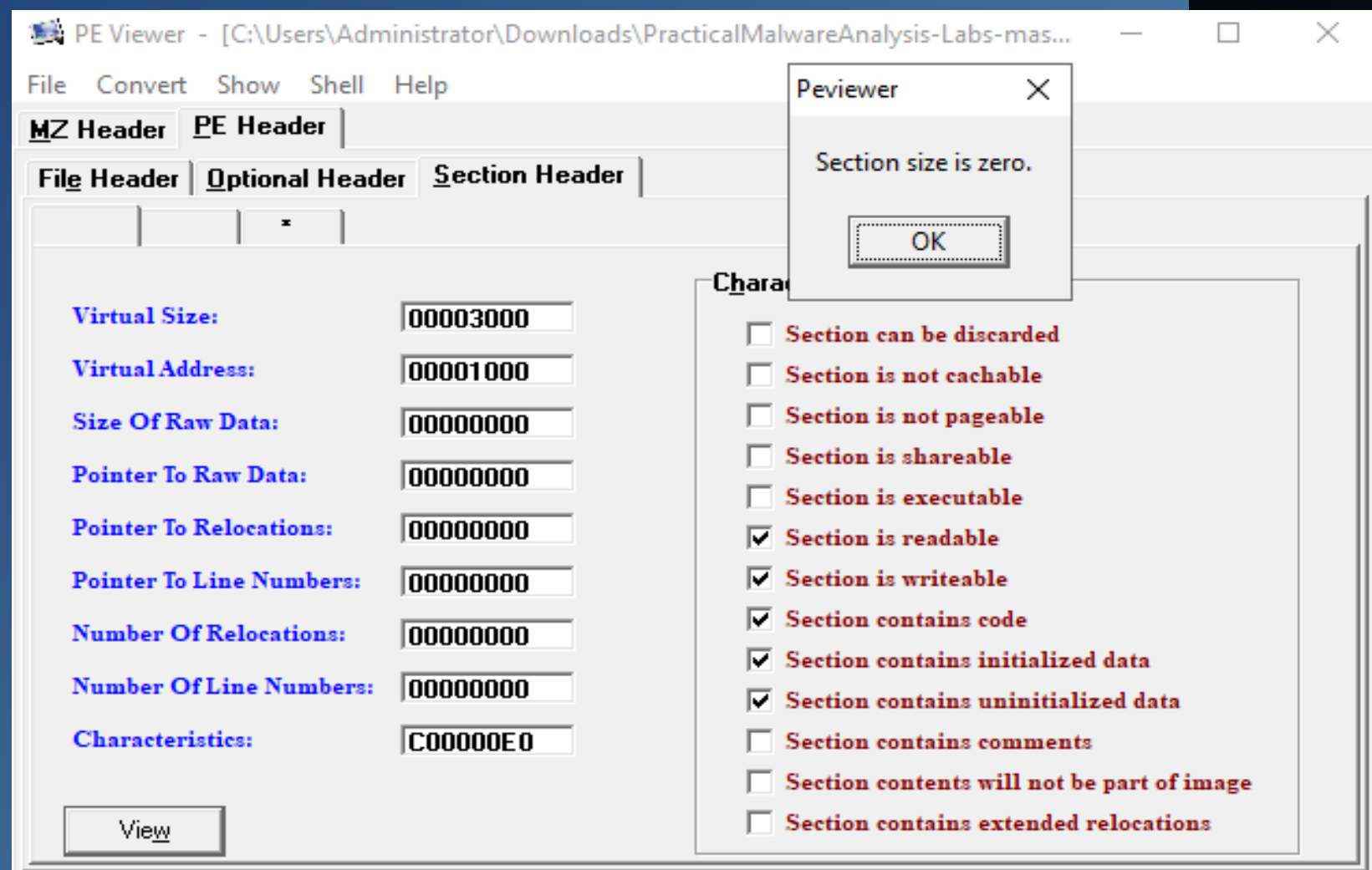
All the screenshots in Step 2 for the sample malware Lab02-04 are the results of using PE Viewer to discover the string values, which are found under Pe-Header-Section Header



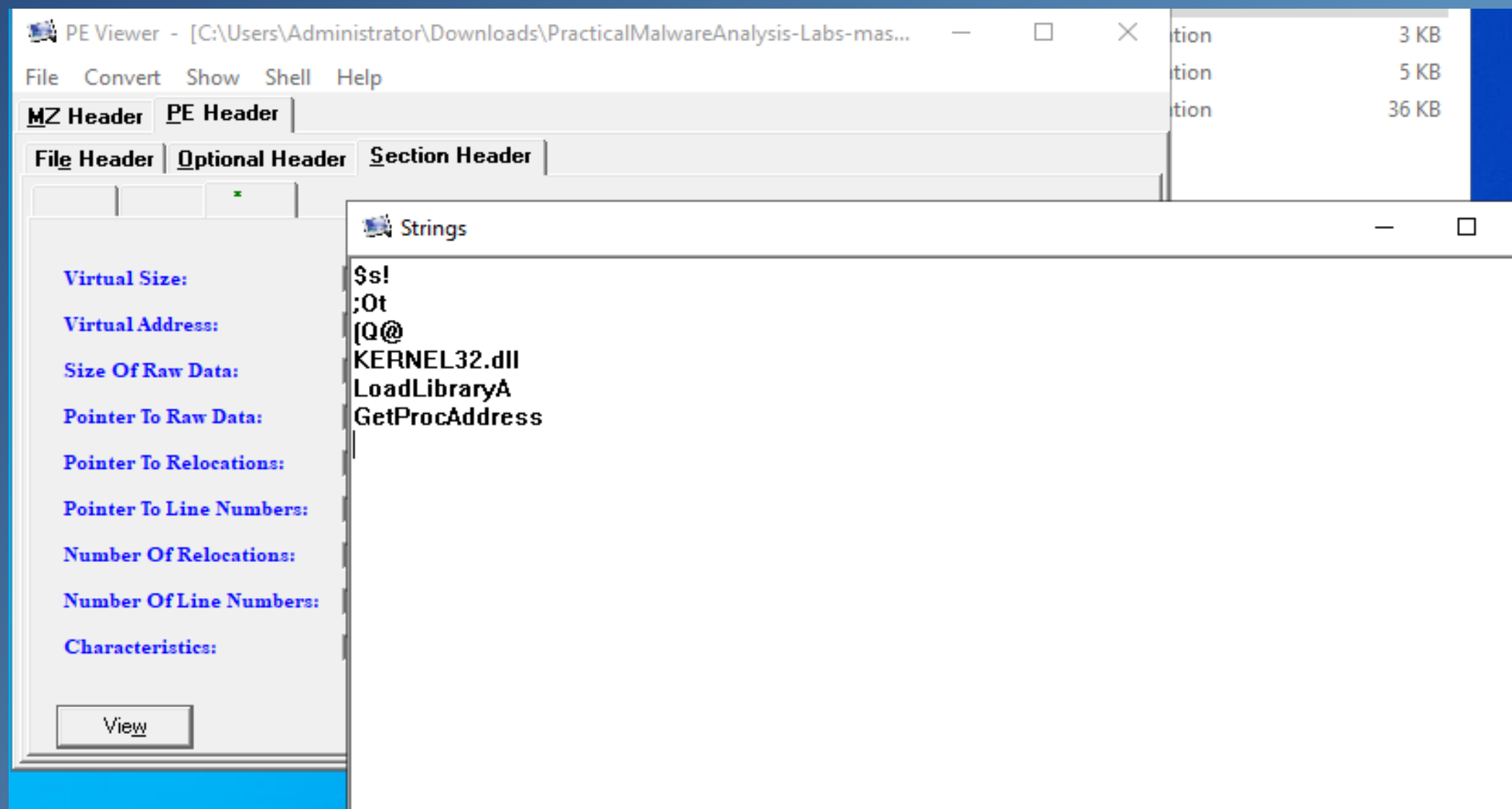
Cont. Step 2: Lab01-02



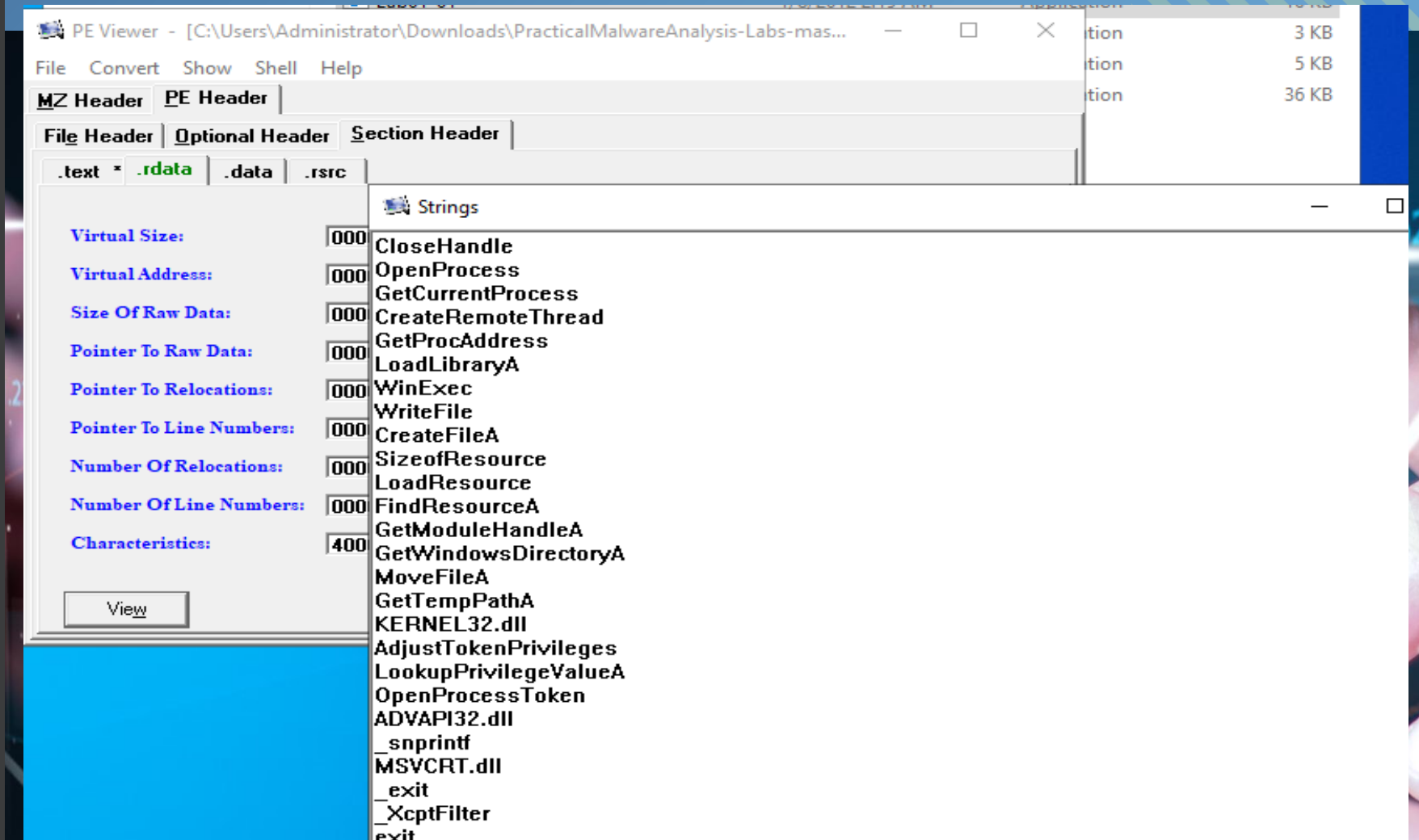
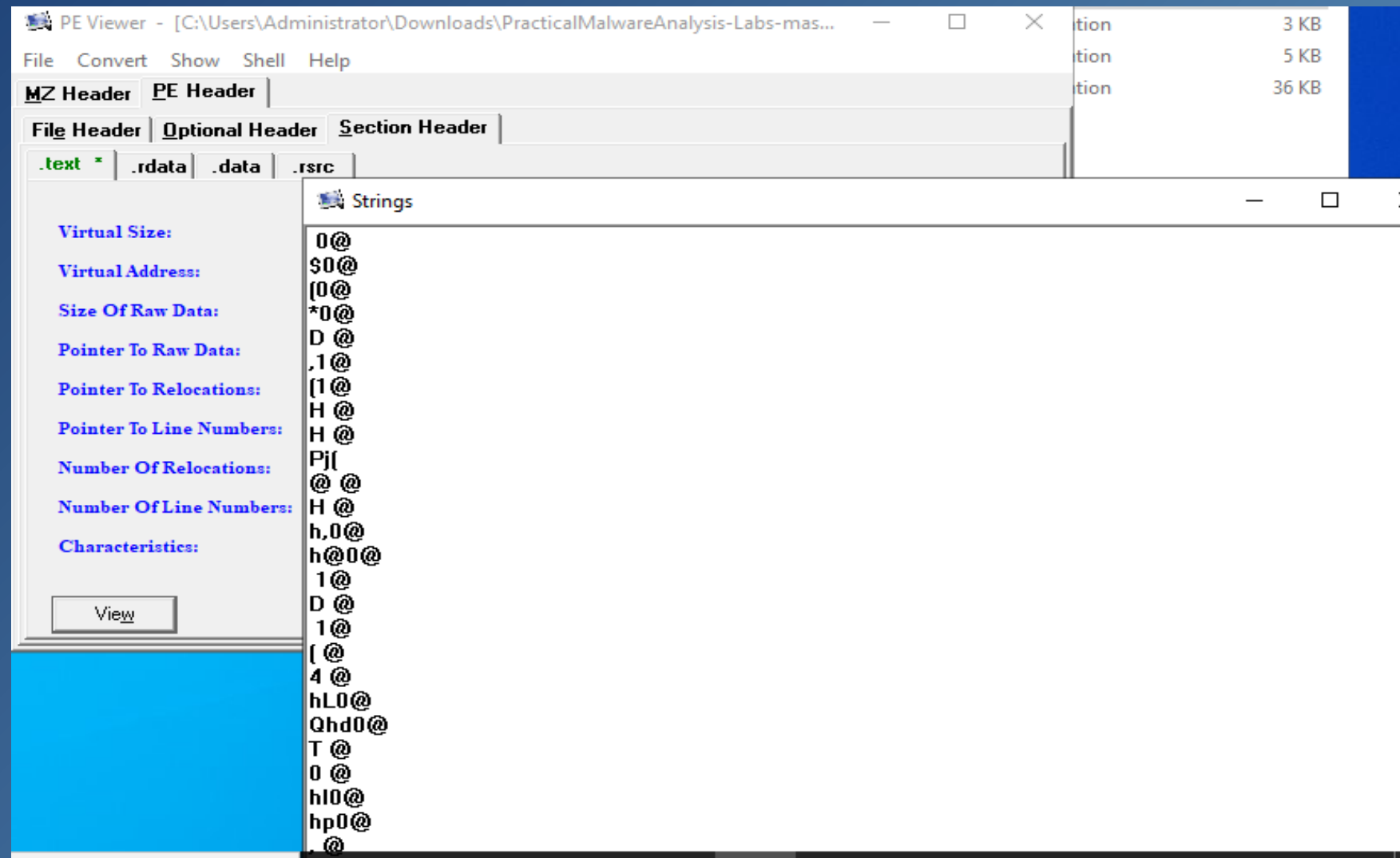
Step 2: Lab01-03



Cont. Step 2: Lab01-03



Step 2: Lab01-04



Cont. Step 2: Lab01-04

PE Viewer - [C:\Users\Administrator\Downloads\PracticalMalwareAnalysis-Labs-mas...]

File Convert Show Shell Help

MZ Header PE Header

File Header Optional Header Section Header

.text *.rdata .data .rsrc

Strings

Virtual Size: 0000014C

Virtual Address: 00003000

Size Of Raw Data: 00001000

Pointer To Raw Data: 00003000

Pointer To Relocations: 00000000

Pointer To Line Numbers: 00000000

Number Of Relocations: 00000000

Number Of Line Numbers: 00000000

Characteristics: C0000004

winlogon.exe

<not real>

SeDebugPrivilege

sfc_os.dll

{system32\wupdmgr.exe

%s%s

BIN

#101

EnumProcessModules

psapi.dll

GetModuleBaseNameA

psapi.dll

EnumProcesses

psapi.dll

{system32\wupdmgr.exe

%s%s

winup.exe

%s%s

View

PE Viewer - [C:\Users\Administrator\Downloads\PracticalMalwareAnalysis-Labs-mas...]

File Convert Show Shell Help

MZ Header PE Header

File Header Optional Header Section Header

.text *.rdata .data .rsrc

Strings

Virtual Size: 00004060

Virtual Address: 00004000

Size Of Raw Data: 00005000

Pointer To Raw Data: 00004000

Pointer To Relocations: 00000000

Pointer To Line Numbers: 00000000

Number Of Relocations: 00000000

Number Of Line Numbers: 00000000

Characteristics: 40000004

GetWindowsDirectoryA

WinExec

GetTempPathA

KERNEL32.dll

URLDownloadToFileA

urlmon.dll

_snprintf

MSVCRT.dll

_exit

_XcptFilter

_exit

_p__initenv

_getmainargs

_initterm

_setusermatherr

_adjust_fdiv

_p__commode

_p__fmode

_set_app_type

_except_handler3

_controlfp

{winup.exe

%s%s

{system32\wupdmgrd.exe

%s%s

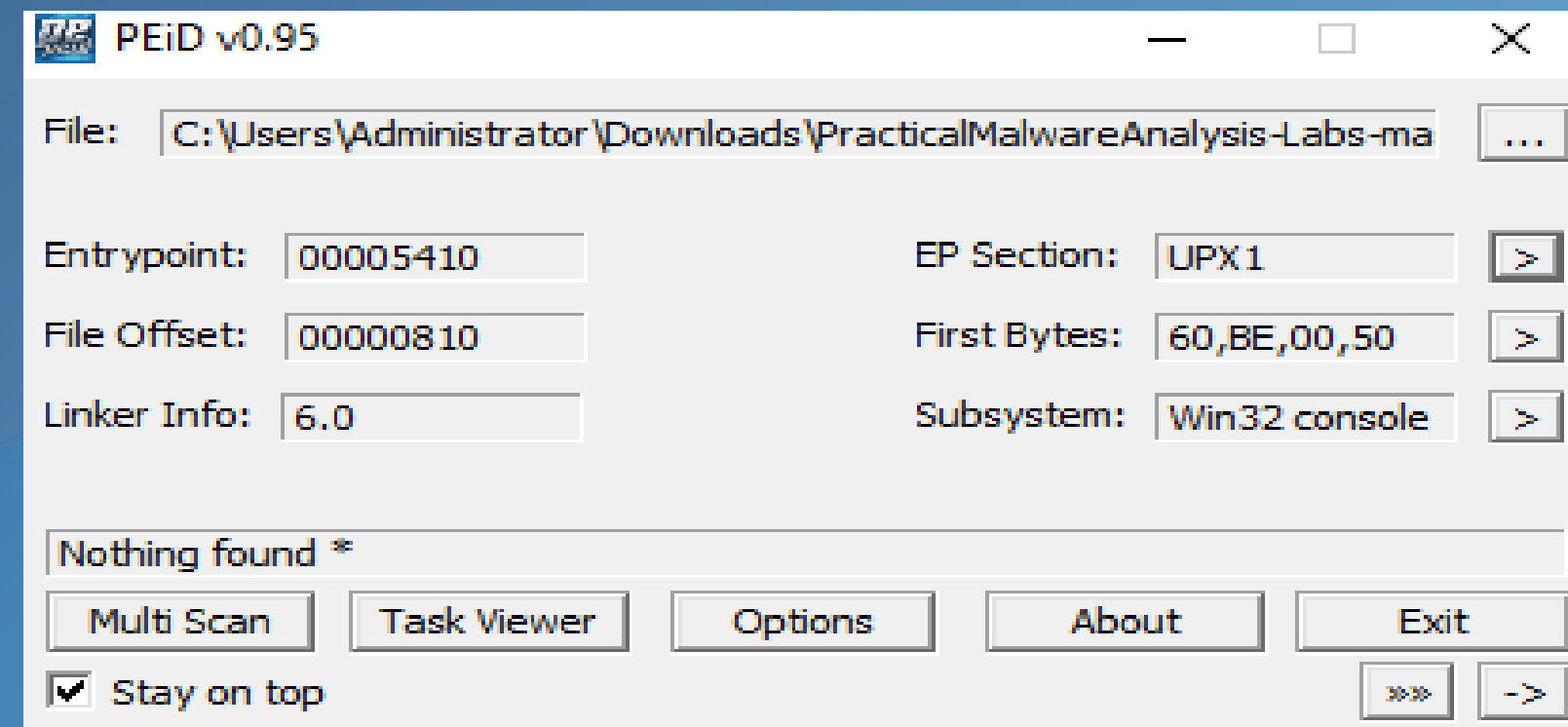
<http://www.practicalmalwareanalysis.com/updater.exe>

View

Step 3: Lab01-02

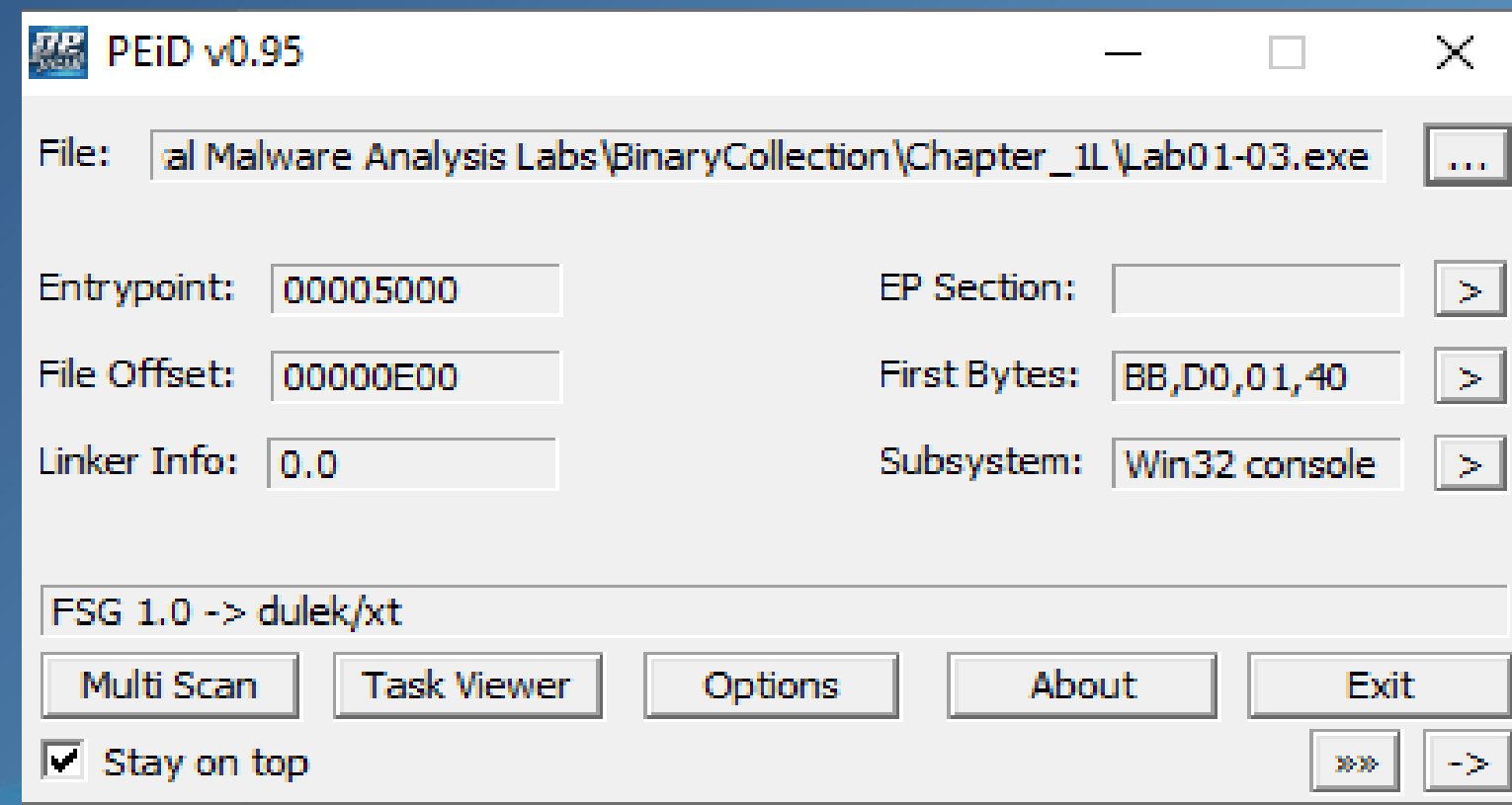
All the screenshots in Step 3 for the sample malware Lab02-04 are the results of using PEiD to find whether the malware was unpacked or not.

The Lab01-02 malware is packed as the UPX1 in EP Section.



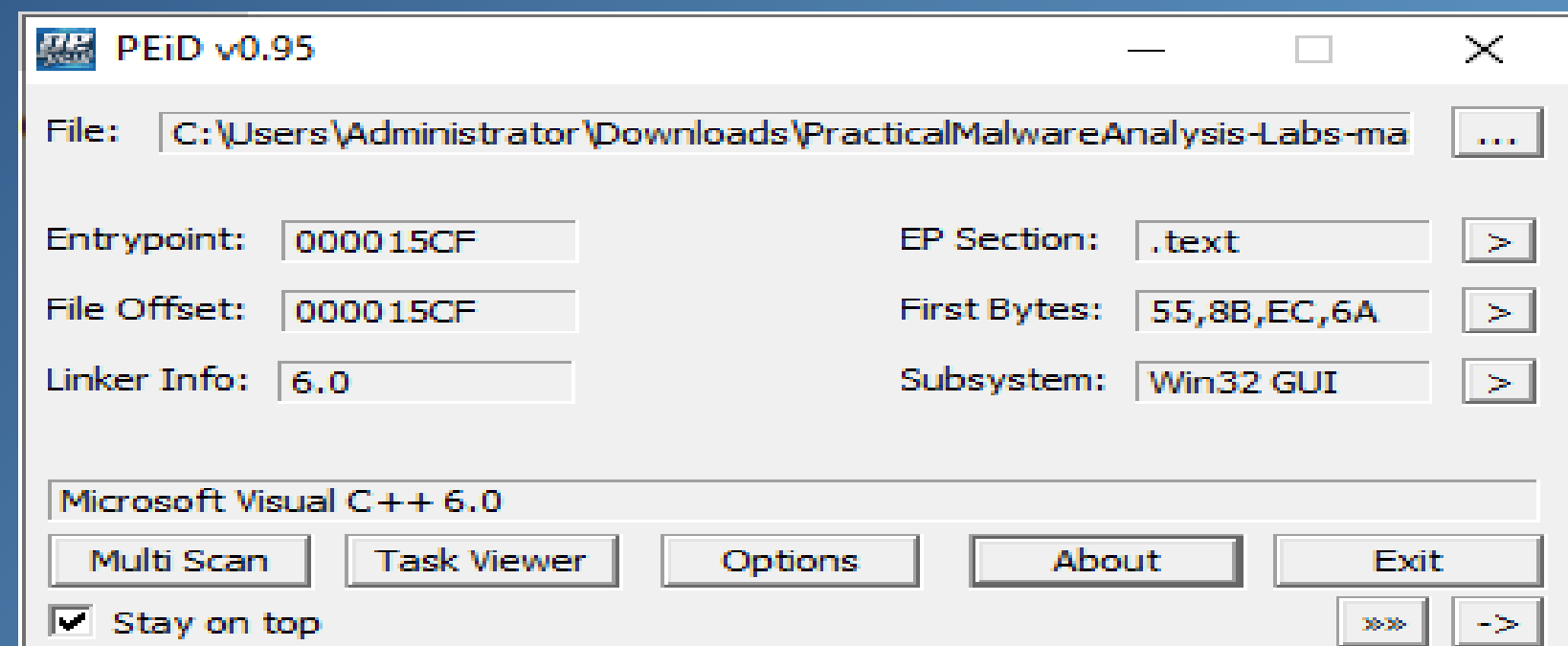
Step 3: Lab01-03

The Lab01-03 malware is corrupted / improperly packed or has a custom Packer on the file as it is blank in the EP Section.



Step 3: Lab01-04

The Lab01-04 malware is unpacked as the .text in EP Section means that the file is still unpacked or no packing was discovered.



Step 4: Lab01-02

PE Explorer - C:\Users\Administrator\Downloads\PracticalMalwareAnalysis-Labs-master\PracticalMalwareAnalysis-Labs-master\PracticalMalwareAnalysis-Labs\P...

File View Tools Help

Address of Entry Point: 00401190 Real Image Checksum: 000041F9h

Field Name	Data Value	Description	Field Name	Data Value	Description
Machine	014Ch	i386®	Section Alignment	00001000h	
Number of Sections	0003h		File Alignment	00001000h	
Time Date Stamp	4D370D01h	19/01/2011 16:10:41	Operating System Version	00000004h	4.0
Pointer to Symbol Table	00000000h		Image Version	00000000h	0.0
Number of Symbols	00000000h		Subsystem Version	00000004h	4.0
Size of Optional Header	00E0h		Win32 Version Value	00000000h	Reserved
Characteristics	010Fh		Size of Image	00004000h	16384 bytes
Magic	010Bh	PE32	Size of Headers	00001000h	
Linker Version	0006h	6.0	Checksum	00000000h	
Size of Code	00001000h		Subsystem	0003h	Win32 Console
Size of Initialized Data	00002000h		Dll Characteristics	0000h	
Size of Uninitialized Data	00000000h		Size of Stack Reserve	00100000h	
Address of Entry Point	00401190h		Size of Stack Commit	00001000h	
Base of Code	00001000h		Size of Heap Reserve	00100000h	
Base of Data	00002000h		Size of Heap Commit	00001000h	
Image Base	00400000h		Loader Flags	00000000h	Obsolete
			Number of Data Directories	00000010h	

```

25.01.2025 13:18:19 : Next Header OFFSET: 00E0h
25.01.2025 13:18:19 : PE Signature: OK
25.01.2025 13:18:19 : Calculating Checksum: SUCCESS (Header's Checksum: 00000000h / Real Checksum: 000041F9h)
25.01.2025 13:18:19 : EOF Position: 00004000h (16384)
25.01.2025 13:18:19 : Done.
  
```

For Help, press F1

The Lab01-02 malware date and implementation

were discovered using the Application PE

Explorer.

Step 4: Lab01-03

PE Explorer - C:\Users\Administrator\Downloads\PracticalMalwareAnalysis-Labs-master\PracticalMalwareAnalysis-Labs-master\PracticalMalwareAnalysis-Labs\P...

File View Tools Help

Address of Entry Point: 00405000 Real Image Checksum: 0000CED2h

Field Name	Data Value	Description	Field Name	Data Value	Description
Machine	014Ch	i386	Section Alignment	00001000h	
Number of Sections	0003h		File Alignment	00001000h	
Time Date Stamp	00000000h	01/01/1970 00:00:00	Operating System Version	00000004h	4.0
Pointer to Symbol Table	00000000h		Image Version	00000000h	0.0
Number of Symbols	00000000h		Subsystem Version	00000004h	4.0
Size of Optional Header	00E0h		Win32 Version Value	00000000h	Reserved
Characteristics	010Fh		Size of Image	00006000h	24576 bytes
Magic	010Bh	PE32	Size of Headers	00001000h	
Linker Version	0000h	0.0	Checksum	00000000h	
Size of Code	00001000h		Subsystem	0003h	Win32 Console
Size of Initialized Data	00002000h		Dll Characteristics	0000h	
Size of Uninitialized Data	00000000h		Size of Stack Reserve	00100000h	
Address of Entry Point	00405000h		Size of Stack Commit	00001000h	
Base of Code	00001000h		Size of Heap Reserve	00100000h	
Base of Data	00002000h		Size of Heap Commit	00001000h	
Image Base	00400000h		Loader Flags	00000000h	Obsolete
			Number of Data Directories	00000010h	

```

25.01.2025 13:25:50 : EOF Extra Data From: 0000128Ch <4748>
25.01.2025 13:25:50 : Length of EOF Extra Data: 00000004h <4> bytes.
25.01.2025 13:25:50 : EOF Position: 00001290h <4752>
25.01.2025 13:25:50 : Warning! Section <> <1> extends beyond the raw file offset of section <> <2>.
25.01.2025 13:25:50 : Done.
  
```

For Help, press F1

The Lab01-03 malware date and implementation were discovered using the Application PE Explorer.

Step 4: Lab01-04

PE Explorer - C:\Users\Administrator\Downloads\PracticalMalwareAnalysis-Labs-master\PracticalMalwareAnalysis-Labs-master\PracticalMalwareAnalysis-Labs\P...

File View Tools Help

HEADERS INFO

Address of Entry Point: 004015CF Real Image Checksum: 0000D3EEh

Field Name	Data Value	Description	Field Name	Data Value	Description
Machine	014Ch	i386®	Section Alignment	00001000h	
Number of Sections	0004h		File Alignment	00001000h	
Time Date Stamp	5D69A2B3h	30/08/2019 22:26:59	Operating System Version	00000004h	4.0
Pointer to Symbol Table	00000000h		Image Version	00000000h	0.0
Number of Symbols	00000000h		Subsystem Version	00000004h	4.0
Size of Optional Header	00E0h		Win32 Version Value	00000000h	Reserved
Characteristics	010Fh		Size of Image	00009000h	36864 bytes
Magic	010Bh	PE32	Size of Headers	00001000h	
Linker Version	0006h	6.0	Checksum	00000000h	
Size of Code	00001000h		Subsystem	0002h	Win32 GUI
Size of Initialized Data	00007000h		Dll Characteristics	0000h	
Size of Uninitialized Data	00000000h		Size of Stack Reserve	00100000h	
Address of Entry Point	004015CFh		Size of Stack Commit	00001000h	
Base of Code	00001000h		Size of Heap Reserve	00100000h	
Base of Data	00002000h		Size of Heap Commit	00001000h	
Image Base	00400000h		Loader Flags	00000000h	Obsolete
			Number of Data Directories	00000010h	

```

25.01.2025 13:30:33 : PE Signature: OK
25.01.2025 13:30:33 : Calculating Checksum: SUCCESS (Header's Checksum: 00000000h / Real Checksum: 0000D3EEh)
25.01.2025 13:30:33 : EOF Position: 00009000h (36864)
25.01.2025 13:30:33 : Precompiling Resources...
25.01.2025 13:30:33 : Done.
  
```

For Help, press F1

The Lab01-04 malware date and
implementation were discovered using the
Application PE Explorer.

Step 5: Lab01-02

PE Explorer Disassembler - <C:\Users\Administrator\Downloads\PracticalMalwareAnalysis-Labs-master\PracticalMalwareAnalysis-Labs-master\PracticalMalware...

File Edit Search View Navigate Help

CODE Z STR P STR LP STR UC STR OFF SET BYTE WORD DOW QW GUID

Address	Disassembly	Comment
00403028	SSZ00403028_HGL345:	
00403028	48474C33343500	db 'HGL345',0
0040302F	00	align 4
00403030	SSZ00403030_http_www_malwareanalysisbook_c:	
00403030	687474703A2F2F777777+	db 'http://www.malwareanalysisbook.co
00403053	00	align 4
00403054	SSZ00403054_Internet_Explorer_8_0:	
00403054	496E7465726E65742045+	db 'Internet Explorer 8.0',0
0040306A	0000	align 4
0040306C	L0040306C:	
0040306C	01000000	dd 00000001h
00403070	L00403070:	
00403070	00	db 00h
00403071	00	db 00h
00403072	00	db 00h
00403073	00	db 00h
00403074	L00403074:	
00403074	00000000	dd 00000000h
00403078	L00403078:	
00403078	00000000	dd 00000000h
0040307C	L0040307C:	
0040307C	00000000	dd 00000000h
00403080	L00403080:	
00403080	00000000	dd 00000000h
00403084	L00403084:	
00403084	00000000	dd 00000000h
00403088	L00403088:	
00403088	00	db 00h
00403089	00	db 00h
0040308A	00	db 00h
0040308B	00	db 00h

Strings View 1 View 2 View 3 View 4

Address	String
00403010:	SSZ00403010_MalService
0040301C:	SSZ0040301C_MalService
00403028:	SSZ00403028_HGL345
00403030:	SSZ00403030_http_www_malwareanalysisbook_c
00403054:	SSZ00403054_Internet_Explorer_8_0

Problem and Messages List:

Name List:

Address	Name
00401120:	L00401120
0040113F:	L0040113F
00401150:	L00401150
0040116D:	L0040116D
00401190:	EntryPoint
00401213:	L00401213
004012A0:	jmp_MSUCRT.dl
004012A6:	jmp_MSUCRT.dl
004012AC:	SUB_L004012AC
004012BE:	L004012BE
004012C1:	SUB_L004012C1
004012D0:	jmp_MSUCRT.dl
004012D6:	jmp_MSUCRT.dl
00402000:	ADVAPI32.dll!
00402004:	ADVAPI32.dll!
00402008:	ADVAPI32.dll!
00402010:	KERNEL32.DLL!
00402014:	KERNEL32.DLL!
00402018:	KERNEL32.DLL!
0040201C:	KERNEL32.DLL!
00402020:	KERNEL32.DLL!
00402024:	KERNEL32.DLL!
00402028:	KERNEL32.DLL!
0040202C:	KERNEL32.DLL!
00402030:	KERNEL32.DLL!
00402038:	MSUCRT.dll!_e
0040203C:	MSUCRT.dll!_X
00402040:	MSUCRT.dll!ex
00402044:	MSUCRT.dll!_
00402048:	MSUCRT.dll!_
0040204C:	MSUCRT.dll!_i
00402050:	MSUCRT.dll!_
00402054:	MSUCRT.dll!_a
00402058:	MSUCRT.dll!_

619 EP: 00403030h Ready ... 00:00:00 13:23:18 25.01.2025

The Lab01-02 malware Disassembling results

Step 5: Lab01-03

PE Explorer Disassembler - <C:\Users\Administrator\Downloads\PracticalMalwareAnalysis-Labs-master\PracticalMalwareAnalysis-Labs-master\PracticalMalware...

File Edit Search View Navigate Help

CODE 2 STR P STR LP STR UC STR OFF SET BYTE WORD DWord GUID

Pointer To RawData: 00000E00h Size Of RawData: 00000200h

```

00405000      EntryPoint:
00405000  4D      dec     ebp
00405001  5A      pop     edx
00405002  90      nop
00405003  0003    add     [ebx],al
00405005  0000    add     [eax],al
00405007  000400  add     [eax+eax],al
0040500A  0000    add     [eax],al
0040500C  FF      db      FFh ; 'n'
0040500D  FF      db      FFh ; 'n'
0040500E  00      db      00h
0040500F  00      db      00h
00405010  B8      db      B8h ; '±'
00405011  00      db      00h
00405012  00      db      00h
00405013  00      db      00h
00405014  00      db      00h
00405015  00      db      00h
00405016  00      db      00h
00405017  00      db      00h
00405018  40      db      40h ; 'e'
00405019  00      db      00h
0040501A  00      db      00h
0040501B  00      db      00h
0040501C  00      db      00h
0040501D  00      db      00h
0040501E  00      db      00h
0040501F  00      db      00h
00405020  00      db      00h
  
```

Problem and Messages List:

Unknown OPCode at: 00405000

Name List:

00405000: EntryPoint

Unprocessed data View 1 View 2 View 3 View 4

```

00404000: [83 1C EC 24 6A 01 FF 15 48 20 40 C0 85 C0 0F 7C ...]
0040500C: [FF FF 00 00 B8 00 00 00 00 00 00 00 40 00 00 00 ...]
00405108: [00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 ...]
  
```

680 EP: 00405000h Ready ... 00:00:00 13:27:21 25.01.2025

The Lab01-03 malware Disassembling results

Step 5: Lab01-04

PE Explorer Disassembler - <C:\Users\Administrator\Downloads\PracticalMalwareAnalysis-Labs-master\PracticalMalwareAnalysis-Labs-master\PracticalMalware...

File Edit Search View Navigate Help

CODE Z STR P STR LP STR UC STR OFF SET BYTE WORD DWORD QWORD GUID

Address	Disassembly	Comment
004015CD	50	push eax
004015CE	C3	retn
EntryPoint:		
004015CF	55	push ebp
004015D0	8BEC	mov ebp,esp
004015D2	6AFF	push FFFFFFFFh
004015D4	6898204000	push L00402098
004015D9	6810174000	push jmp_MSUCRT.dll!_except_handler3
004015DE	64A100000000	mov eax,fs:[00000000h]
004015E4	50	push eax
004015E5	64892500000000	mov fs:[00000000h],esp
004015EC	83EC20	sub esp,00000020h
004015EF	53	push ebx
004015F0	56	push esi
004015F1	57	push edi
004015F2	8965E8	mov [ebp-18h],esp
004015F5	8365FC00	and dword ptr [ebp-04h],00000000h
004015F9	6A01	push 00000001h
004015FB	FF1580204000	call [MSUCRT.dll!__set_app_type]
00401601	59	pop ecx
00401602	830D40314000FF	or dword ptr [L00403140],FFFFFFFFh
00401609	830D44314000FF	or dword ptr [L00403144],FFFFFFFFh
00401610	FF157C204000	call [MSUCRT.dll!__p__fmode]
00401616	8B0D3C314000	mov ecx,[L0040313C]
0040161C	8908	mov [eax],ecx
0040161E	FF1578204000	call [MSUCRT.dll!__p__commode]
00401624	8B0D38314000	mov ecx,[L00403138]
0040162A	8908	mov [eax],ecx
0040162C	A174204000	mov eax,[MSUCRT.dll!_adjust_fdiv]
00401631	8B00	mov eax,[eax]
00401633	A348314000	mov [L00403148],eax

Problem and Messages List:

Name List:

00401000:	SUB_L00401000
004010C2:	L004010C2
004010EB:	L004010EB
004010F7:	L004010F7
004010FC:	SUB_L004010FC
00401120:	L00401120
00401153:	L00401153
0040116E:	L0040116E
00401174:	SUB_L00401174
004011A1:	L004011A1
004011D8:	L004011D8
004011F8:	L004011F8
004011FC:	SUB_L004011FC
00401350:	SUB_L00401350
00401419:	L00401419
00401423:	L00401423
0040144A:	L0040144A
00401465:	L00401465
00401474:	L00401474
004014CF:	L004014CF
004014D1:	L004014D1
004014E4:	L004014E4
00401593:	L00401593
00401598:	L00401598
004015A0:	SUB_L004015A0
004015AC:	L004015AC
004015C0:	L004015C0
004015CF:	EntryPoint
00401652:	L00401652
004016E0:	jmp_MSUCRT.dll
004016E6:	jmp_MSUCRT.dll
004016EC:	SUB_L004016EC
004016FE:	L004016FE
00401701:	SUB_L00401701

Strings View 1 View 2 View 3 View 4

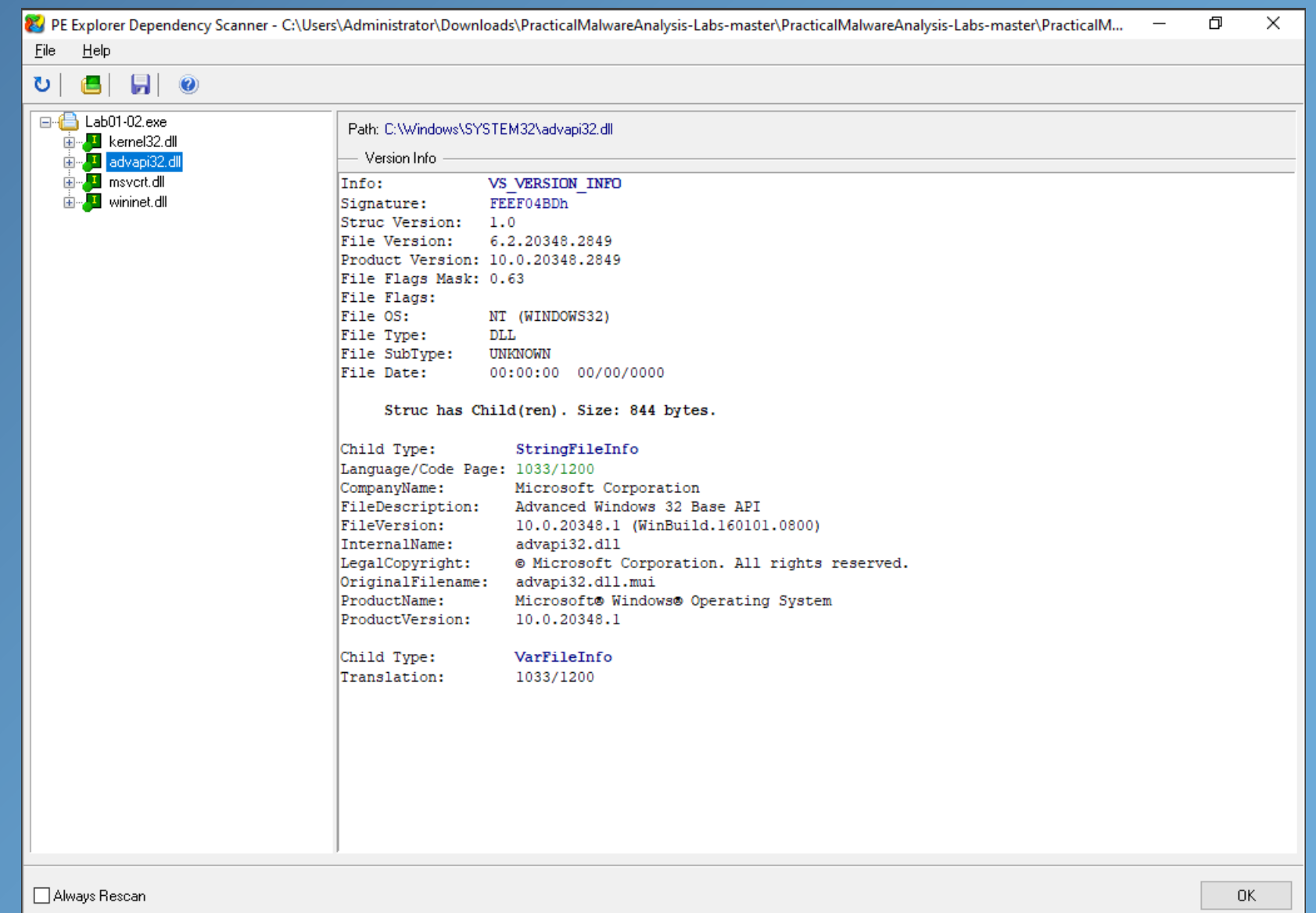
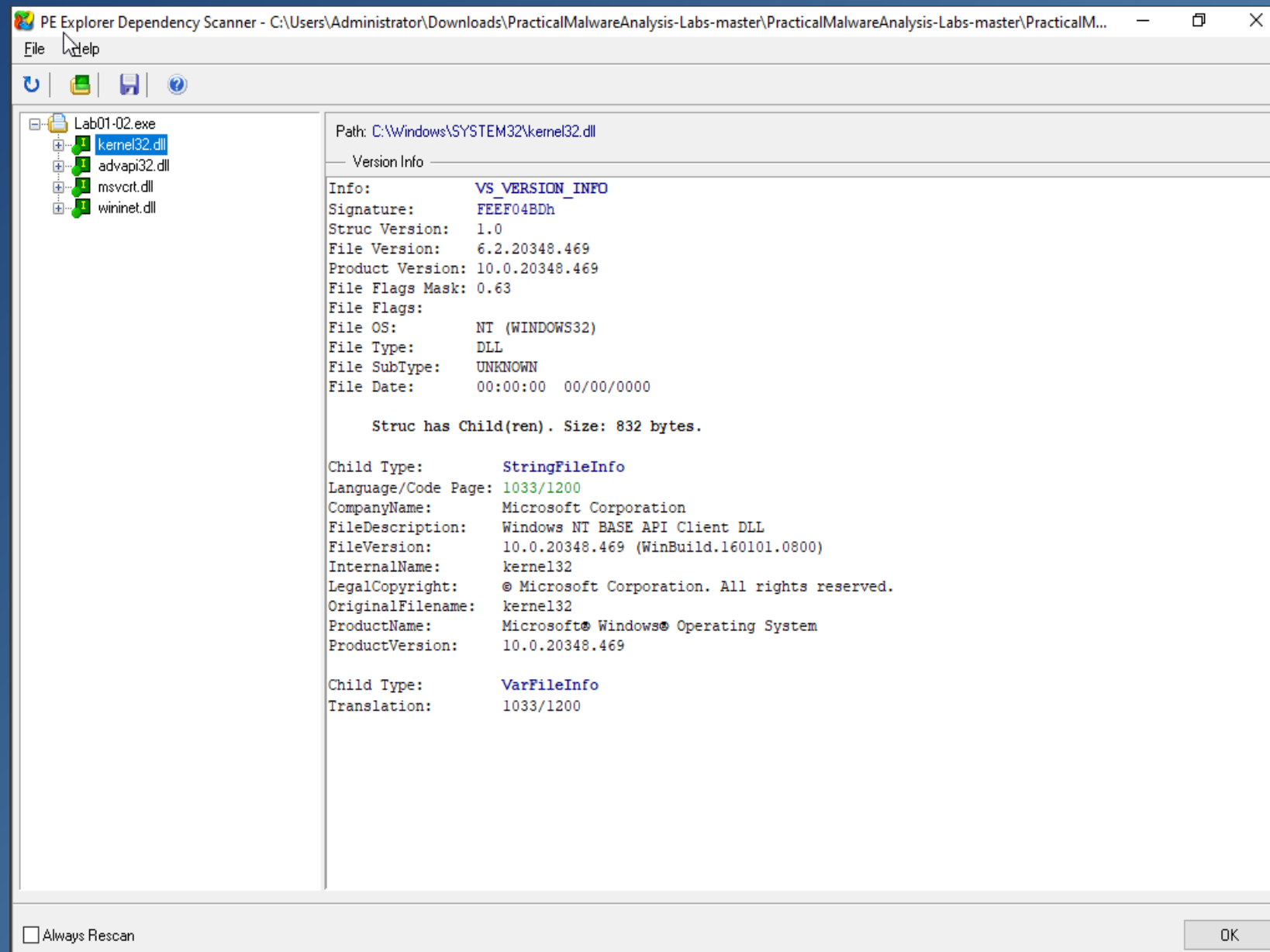
0040302C:	SSZ0040302C_SeDebugPrivilege	'SeDebugPrivilege',0
00403040:	SSZ00403040_sfc_os_dll	'sfc_os.dll',0
0040304C:	SSZ0040304C__system32_wupdmgr_exe	'\system32\wupdmgr.exe',0
00403064:	SSZ00403064__s_s	'%s%s',0
00403070:	SSZ00403070__101	'#101',0
00403078:	SSZ00403078_EnumProcessModules	'EnumProcessModules',0
0040308C:	SSZ0040308C_psapi_dll	'psapi.dll',0
00403098:	SSZ00403098_GetModuleBaseNameA	'GetModuleBaseNameA',0
004030AC:	SSZ004030AC_psapi_dll	'psapi.dll',0

441 EP: 004015CFh Ready ... 00:00:00 13:31:59 25.01.2025

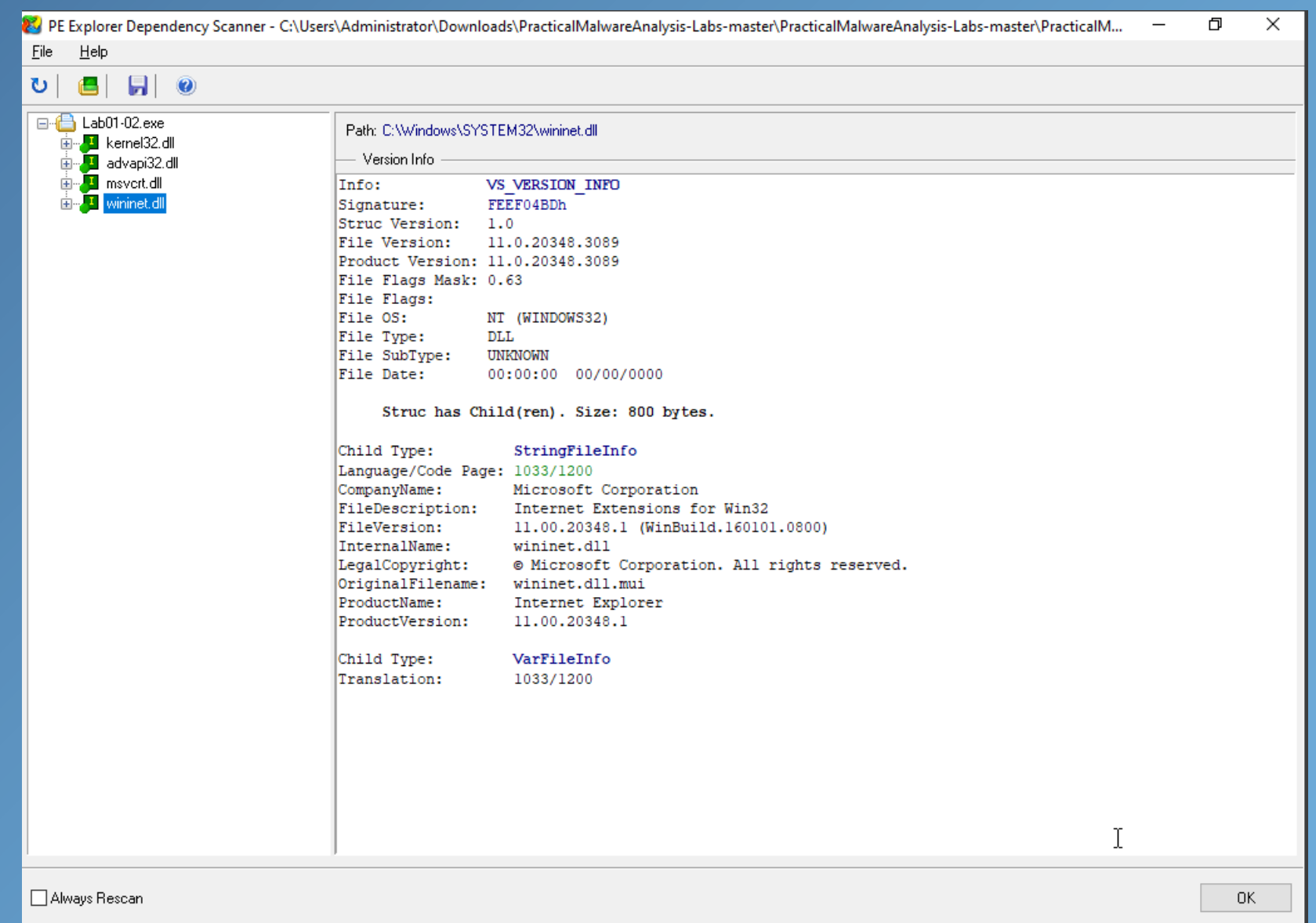
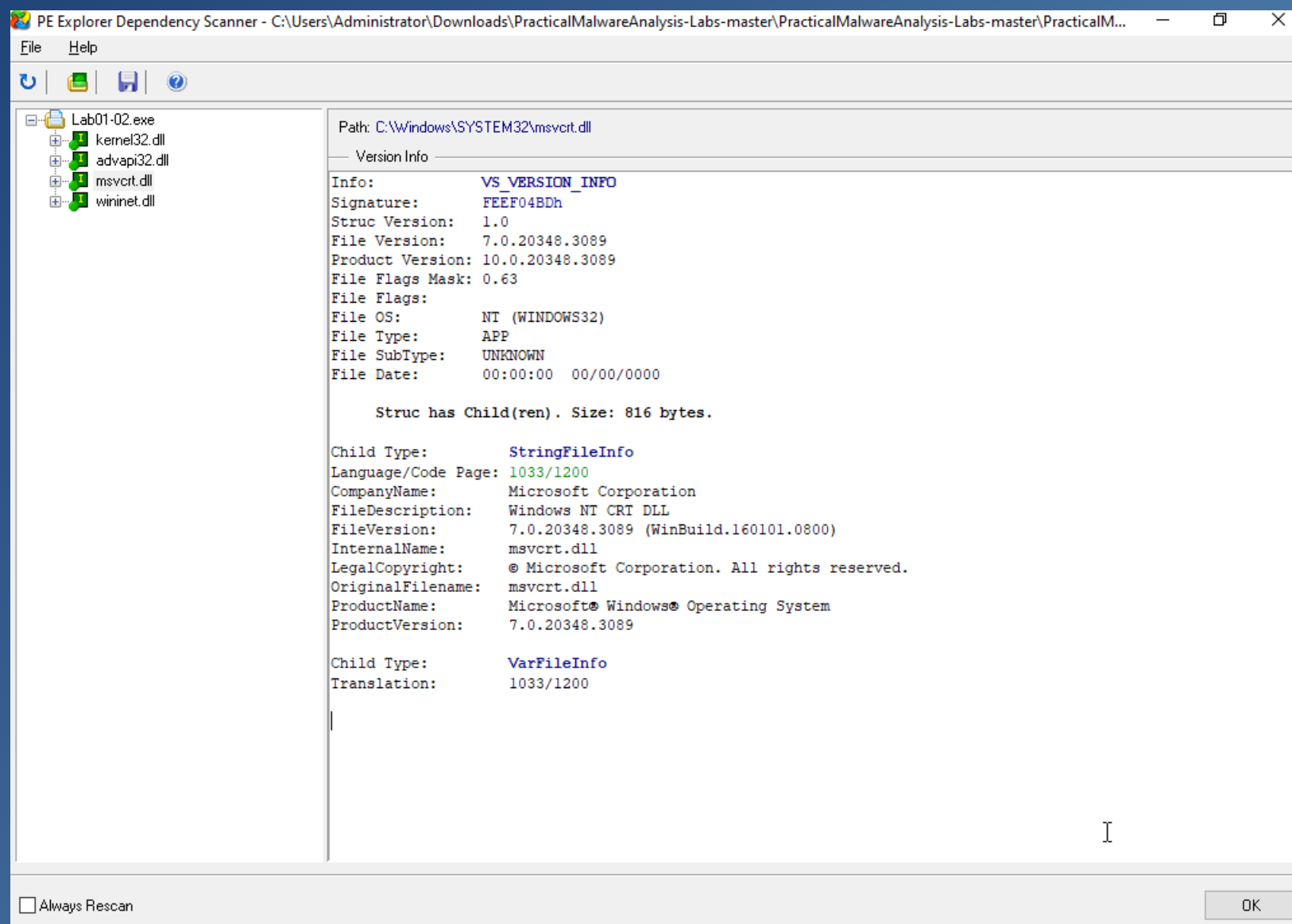
The Lab01-04 malware Disassembling results

Step 6: Lab01-02

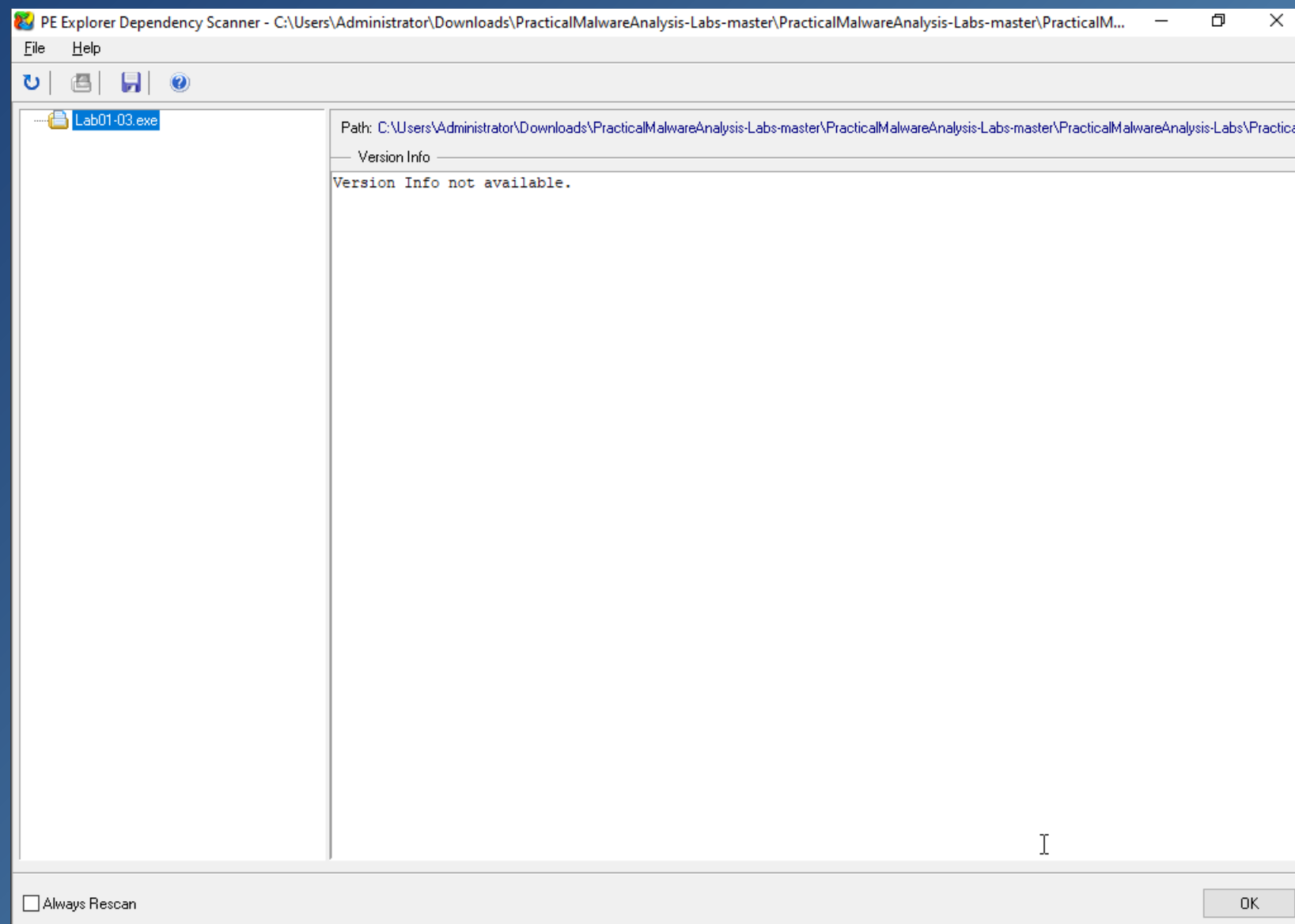
The Lab01-02 malware import functions are shown below



Cont. Step 6: Lab01-02



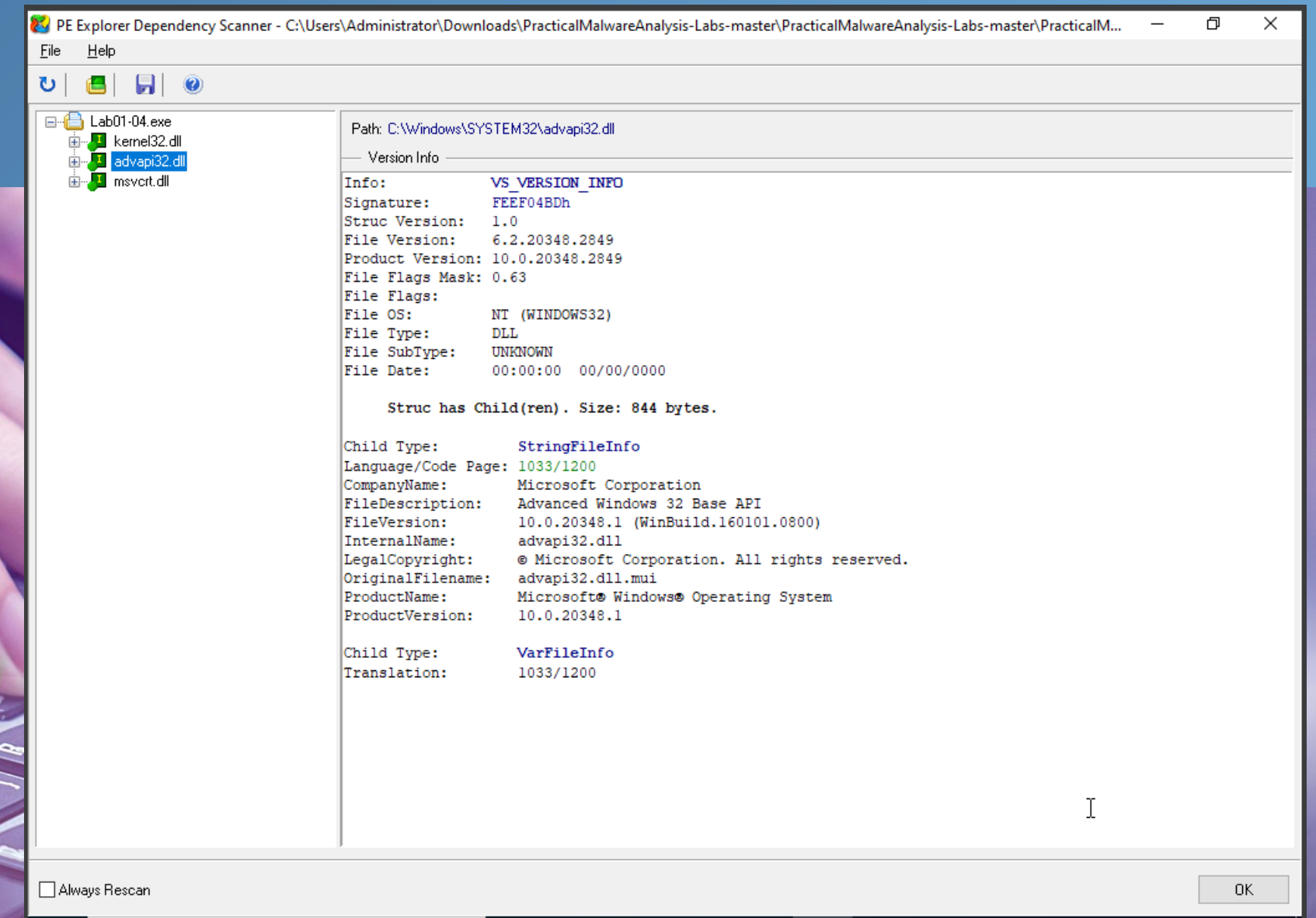
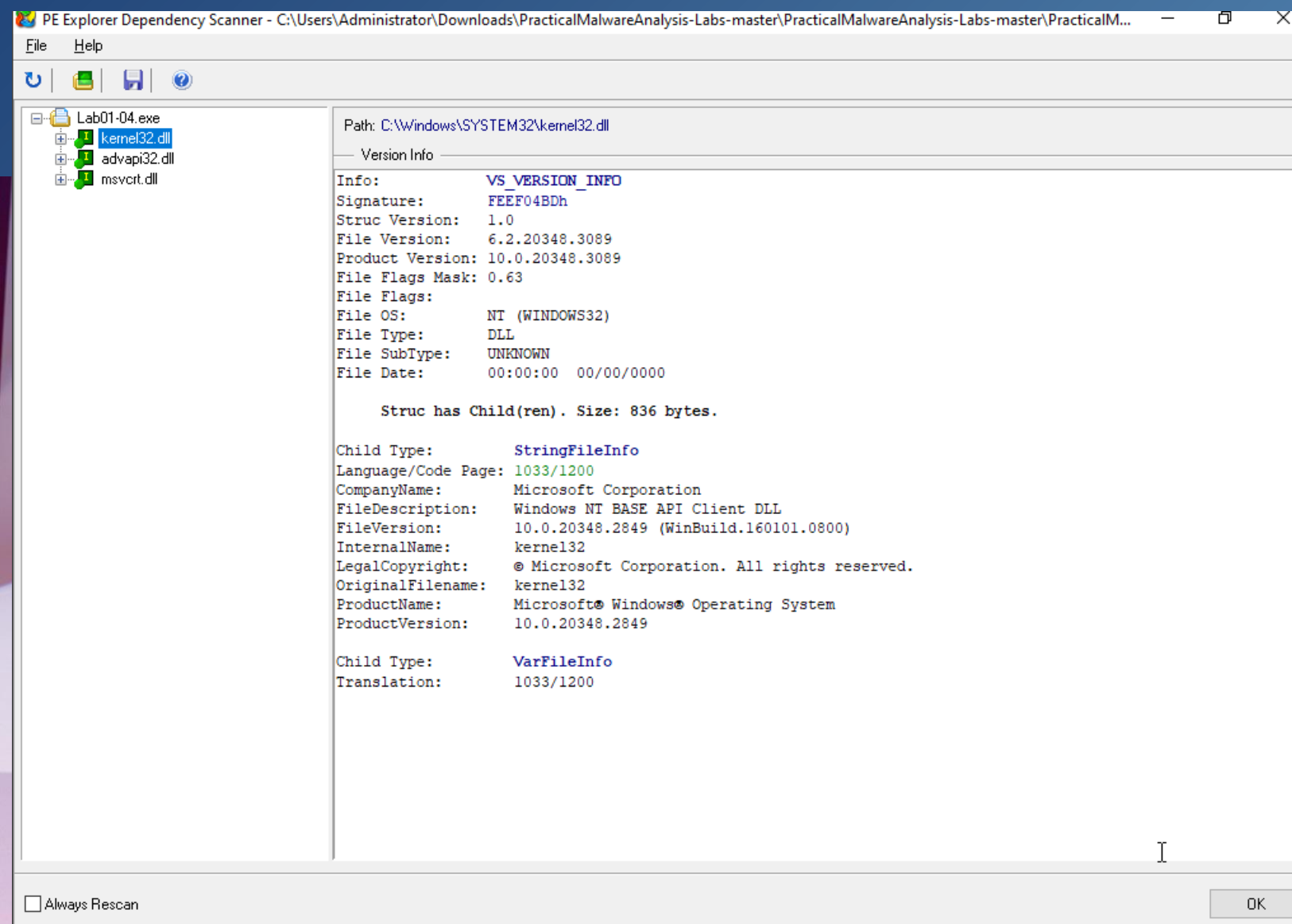
Step 6: Lab01-03



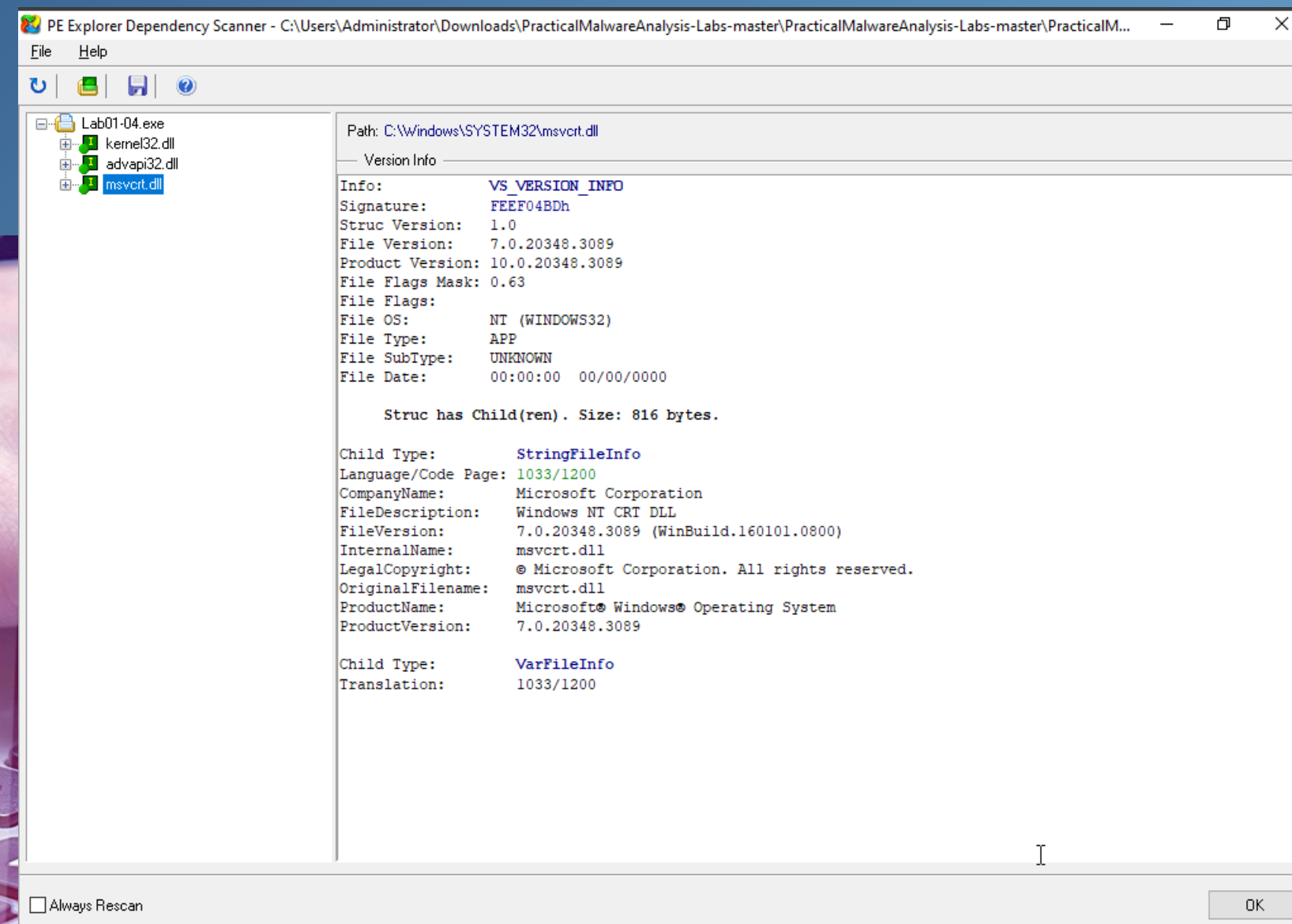
The Lab01-03 malware doesn't have any import functions attached to the file.

Step 6: Lab01-04

The Lab01-04 malware import functions are shown below



Cont. Step 6: Lab01-04



CONCLUSION

- In conclusion the project provides a structured and methodical approach to understanding malware through static analysis. By leveraging tools such as VirusTotal, PEViewer, PEiD, and PE-Explorer, the analysis process uncovered critical details about sample malware, including its packing status, metadata, string values, and import functions.
- The project highlighted the importance of thorough inspection and effective tool utilization in malware analysis, ensuring ethical and secure practices. It underscores the need for continuous learning in cybersecurity to counter evolving malware threats and protect digital systems

References

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THANK YOU